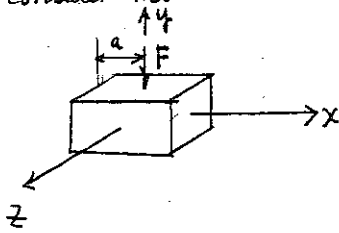


CHAPTER 5

(5-1) consider the elastic material block



the block is just allowed to deform in the y-direction

$$\therefore \epsilon_y = \Delta \quad \epsilon_x = \epsilon_z = 0$$

From stress-strain relationship

$$\sigma_x - \nu(\sigma_y + \sigma_z) = E\epsilon_x = 0 \quad (1)$$

$$\sigma_y - \nu(\sigma_x + \sigma_z) = E\epsilon_y = E\Delta \quad (2)$$

$$\sigma_z - \nu(\sigma_x + \sigma_y) = E\epsilon_z = 0 \quad (3)$$

From (1) & (3)

$$\sigma_x(1+\nu) - \sigma_z(1+\nu) = 0 \quad \therefore \sigma_x = \sigma_z$$

From (2)

$$\sigma_y - \nu(2\sigma_x) = E\Delta \quad \rightarrow \quad \sigma_x = -\frac{E\Delta}{2\nu} + \frac{\sigma_y}{2\nu} \quad (4)$$

Putting (4) into (3) and using $\sigma_x = \sigma_z$

$$\sigma_x(1-\nu) - \nu\sigma_y = 0 \quad \rightarrow \quad (1-\nu)\left[-\frac{E\Delta}{2\nu} + \frac{\sigma_y}{2\nu}\right] = \nu\sigma_y$$

Solving for σ_y :

$$\sigma_y \left[\nu - \frac{1-\nu}{2\nu} \right] = -\frac{1-\nu}{2\nu} E\Delta$$

$$\sigma_y(1+\nu)(2\nu-1) = -(1-\nu)E\Delta$$

$$\therefore \sigma_y = \frac{(1-\nu)}{(1+\nu)(1-2\nu)} E\Delta$$

Putting in terms of F_0, a, L, C ,

$$\frac{F_0}{4a^2} = \frac{(1-\nu)}{(1+\nu)(1-2\nu)} \frac{E C}{L} \quad \rightarrow \quad F_0 = \frac{(1-\nu)}{(1+\nu)(1-2\nu)} \frac{4ECa^2}{L}$$

5.2

Volume of parallelepiped before strain

$$= (\Delta x)(\Delta y)(\Delta z) = V$$

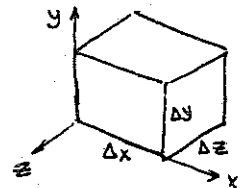
After strain the lengths Δx etc

become: $(\Delta x + \epsilon_x \Delta x); (\Delta y + \epsilon_y \Delta y); (\Delta z + \epsilon_z \Delta z)$

$$\Delta V = (\Delta x + \epsilon_x \Delta x)(\Delta y + \epsilon_y \Delta y)(\Delta z + \epsilon_z \Delta z) - (\Delta x)(\Delta y)(\Delta z)$$

$$= (\epsilon_x + \epsilon_y + \epsilon_z)(\Delta x \Delta y \Delta z) \text{ retaining only first order quantities}$$

$$\therefore \frac{\text{Vol. change}}{\text{original volume}} = \frac{\Delta V}{V} = \epsilon_x + \epsilon_y + \epsilon_z$$



5.3 a) $\Delta V/V = \epsilon_x + \epsilon_y + \epsilon_z$ (prob 5.2)

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)]$$

$$\epsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_z + \sigma_x)] \quad (\text{constant temperature})$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)]$$

$$\therefore \frac{\Delta V}{V} = \frac{1}{E} [\sigma_x + \sigma_y + \sigma_z - 2\nu(\sigma_x + \sigma_y + \sigma_z)] = (\sigma_x + \sigma_y + \sigma_z) \frac{(1-2\nu)}{E}$$

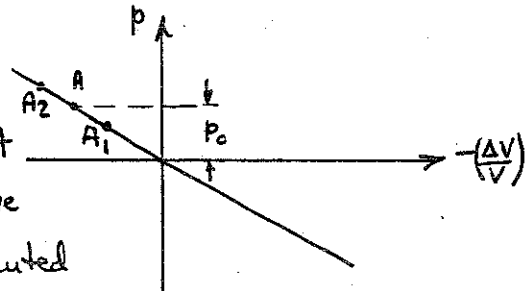
b) Under hydrostatic pressure p ,

$$\sigma_x = \sigma_y = \sigma_z = -p$$

$$\frac{\Delta V}{V} = - \frac{3p(1-2\nu)}{E} \quad \text{and} \quad B \equiv \frac{p}{-(\frac{\Delta V}{V})} = \frac{E}{3(1-2\nu)}$$

5.4 A material with $\nu > \frac{1}{2}$ will have $B < 0$ and the pressure vs volume change curve shown below:

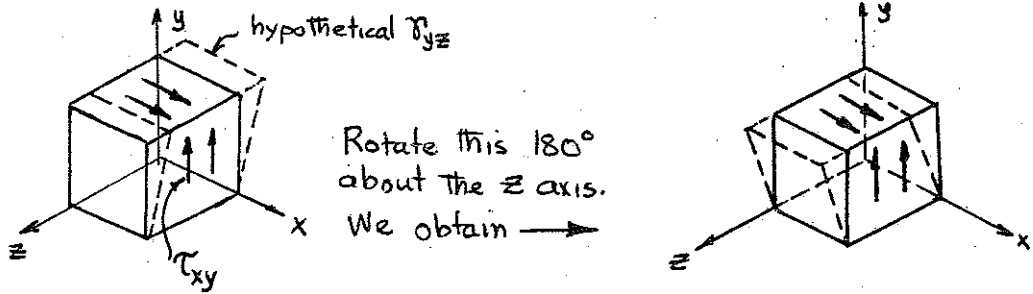
Suppose a piece of material is at equilibrium in a constant pressure environment at pressure P_0 (see figure). The state of the material is represented by point A.



Now let us suppose that due to some fluctuation (eg thermal) the volume of the material increases by a small amount. The material passes to state A_2 . The pressure in the material is raised, and there is now a pressure differential which will accelerate the material in the direction of volume increase. Eventually, the volume will go to infinity.

Similarly, a small volume decrease will cause a pressure drop within the material, and the volume decrease will accelerate and eventually lead to zero volume. Such materials are therefore unstable.

5.5

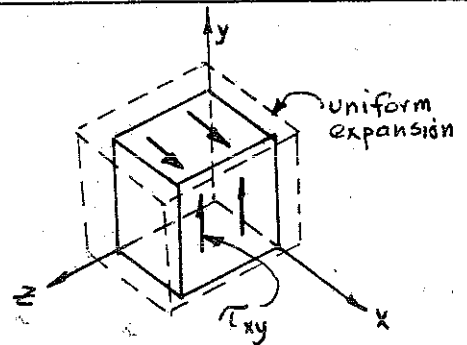


After rotation: τ_{xy} is unchanged but τ_{yz} is reversed. Hence τ_{yz} must be zero because if it were not then its sign would depend on the orientation of the coordinate axes which contradicts the postulated isotropy.

5.6

Rotate 180° about y axis. Shear stress is reversed but not uniform expansion. Assuming that the expansion (or contraction) and the shear stress are linearly related

thus: (expansion) = $C(\tau_{xy})$, then C must be zero and the expansion is also zero.



(5.7) Eq(5.8): $\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] + \alpha(T - T_0)$; $\epsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)] + \alpha(T - T_0)$
 $\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)] + \alpha(T - T_0)$
 $\gamma_{xy} = \frac{\tau_{xy}}{G}$; $\gamma_{yz} = \frac{\tau_{yz}}{G}$; $\gamma_{zx} = \frac{\tau_{zx}}{G}$

$$\left. \begin{aligned} \sigma_x - \nu\sigma_y - \nu\sigma_z &= E[\epsilon_x - \alpha(T - T_0)] \\ -\nu\sigma_x + \sigma_y - \nu\sigma_z &= E[\epsilon_y - \alpha(T - T_0)] \\ -\nu\sigma_x - \nu\sigma_y + \sigma_z &= E[\epsilon_z - \alpha(T - T_0)] \end{aligned} \right\} \begin{bmatrix} 1 & -\nu & -\nu \\ -\nu & 1 & -\nu \\ -\nu & -\nu & 1 \end{bmatrix} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \end{bmatrix} = E \begin{bmatrix} \epsilon_x - \alpha(T - T_0) \\ \epsilon_y - \alpha(T - T_0) \\ \epsilon_z - \alpha(T - T_0) \end{bmatrix}$$

From the matrix equation,

$$\sigma_x = \frac{[(1-\nu)E\epsilon_x + \nu E(\epsilon_y + \epsilon_z) + (1-\nu)\alpha(T - T_0)]}{(1+\nu)(1-2\nu)}$$

$$\sigma_y = \frac{[(1-\nu)E\epsilon_y + \nu E(\epsilon_x + \epsilon_z) + (1-\nu)\alpha(T - T_0)]}{(1+\nu)(1-2\nu)}$$

$$\sigma_z = \frac{[(1-\nu)E\epsilon_z + \nu E(\epsilon_x + \epsilon_y) + (1-\nu)\alpha(T - T_0)]}{(1+\nu)(1-2\nu)}$$

$$\tau_{xy} = G\gamma_{xy}; \quad \tau_{yz} = G\gamma_{yz}; \quad \tau_{zx} = G\gamma_{zx}$$

-8 $\sigma_x = 130$ $E = 205 \text{ GN/m}^2$

$\sigma_y = -70$ $\nu = 0.3$

$\tau_{xy} = 80$

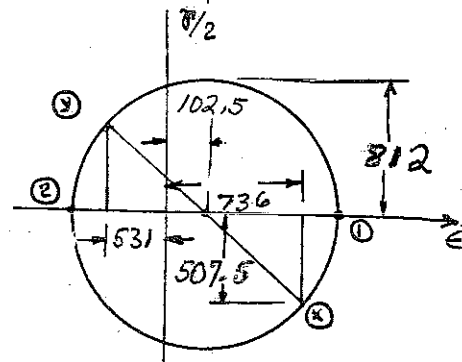
$$\therefore \epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] = 736 \times 10^{-6}$$

$$\epsilon_y = -531 \times 10^{-6}$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G} = \frac{2(1+\nu)\tau_{xy}}{E} = 1015 \times 10^{-6}$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)] = -87 \times 10^{-6}$$

Mohr's Circle (strains)
(strain $\times 10^6$ is plotted)



$$\epsilon_1 = 914 \times 10^{-6}$$

$$\epsilon_2 = -709 \times 10^{-6}$$

5.9

$\sigma_x = 145 \text{ MN/m}^2$; $\tau_{xy} = 42 \text{ MN/m}^2$; $\epsilon_z = -3.6 \times 10^{-4}$

$\sigma_z = 0$ $\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)]$

$\sigma_y = 95 \text{ MN/m}^2$

5.10

$$\epsilon_1 = 3.2 \times 10^{-4} ; \epsilon_2 = -5.4 \times 10^{-4}$$

We will first find σ_1 and σ_2 and use Mohr's circle to find $\sigma_x, \sigma_y, \tau_{xy}$.

Since $\sigma_3 = 0$,

$$\epsilon_1 = \frac{1}{E} [\sigma_1 - \nu \sigma_2] ; \epsilon_2 = \frac{1}{E} [\sigma_2 - \nu \sigma_1]$$

Solving these equations for σ_1 and σ_2 :

$$\sigma_1 = \frac{E}{1-\nu^2} [\epsilon_1 + \nu \epsilon_2] ; \sigma_2 = \frac{E}{1-\nu^2} [\epsilon_2 + \nu \epsilon_1]$$

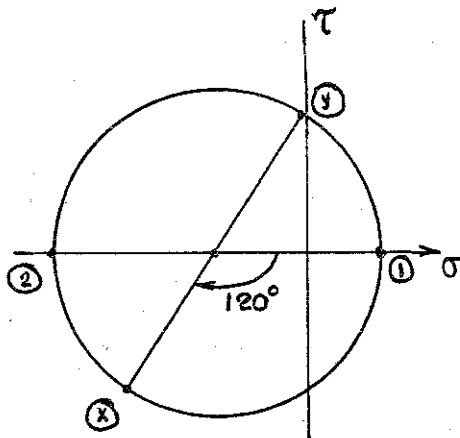
From table 5.1 : $E = 11 \times 10^6$ psi ; $\nu = 0.33$:

$$E/(1-\nu^2) = 11 \times 10^6 / (1 - 0.109) = 12.3 \times 10^6$$

$$\therefore \sigma_1 = (12.3 \times 10^6) [3.2 - 0.33(5.4)] 10^{-4} = 1750 \text{ psi}$$

$$\sigma_2 = (12.3 \times 10^6) [-5.4 + 0.33(3.2)] 10^{-4} = -5350 \text{ psi}$$

MOHR'S CIRCLE :



$$\sigma_x = -3585 \text{ psi} = -24.72 \text{ MN/m}^2$$

$$\sigma_y = -25 \text{ psi} = 0.17 \text{ MN/m}^2$$

$$\tau_{xy} = 3070 \text{ psi} = 21.17 \text{ MN/m}^2$$

5.11

$$\epsilon = \ln \left(\frac{A_0}{A} \right) \quad \therefore \frac{A}{A_0} = e^{-\epsilon}$$

$$\text{Reduction in area} = \frac{A_0 - A}{A_0} = 1 - \frac{A}{A_0} = 1 - e^{-\epsilon}$$

5.12 From prob 4.12:

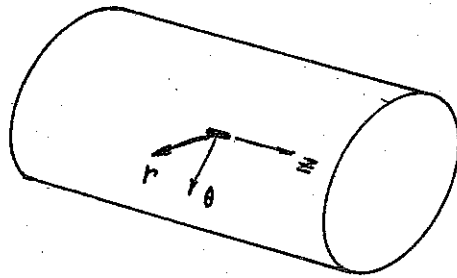
$$\sigma_{\theta} = \frac{pr}{t}$$

$$\sigma_z = \frac{pr}{2t}$$

and $\sigma_r = 0$ at the outside.

$$\epsilon_z = \epsilon_0 = \frac{1}{E} [\sigma_z - \nu(\sigma_r + \sigma_{\theta})]$$

$$= \frac{1}{E} \left[\frac{pr}{2t} - \nu \frac{pr}{t} \right] = \frac{pr}{2Et} (1-2\nu) \quad \therefore p = \frac{2E\epsilon_0 t}{r(1-2\nu)}$$



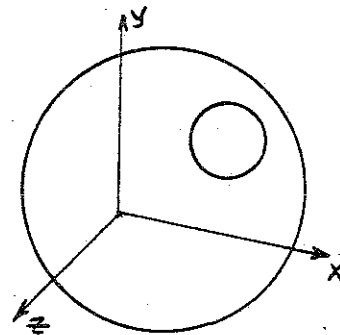
5.13 After Heating, the plate will be stress free with uniform strains:

$$\epsilon_x = \epsilon_y = \epsilon_z = \alpha(T - T_0)$$

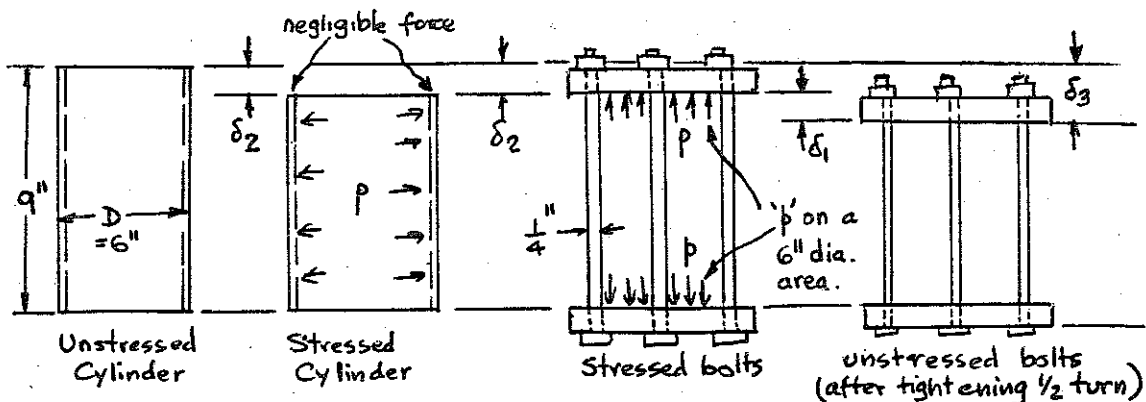
$$\gamma_{xy} = \gamma_{yz} = \gamma_{zx} = 0$$

(This solution satisfies equilibrium, compatibility, stress-strain-temperature relations, boundary conditions.)

Thus the displacement consists only of a uniform change of scale, and the circles will remain circles but their sizes will increase.



5.14



The vessel is certain to leak when the pressure elongates the bolts and shrinks the cylinder by an amount just sufficient to prevent contact between the end plates and the cylinder. At this condition the axial force on the cylinder will be negligible.

GEOMETRY : $\delta_1 + \delta_2 = \delta_3 = \frac{1}{2}(\frac{1}{16}) = 0.025''$ ——— (1)

EQUILIBRIUM : Cylinder : $\sigma_\theta = p \frac{D}{2t} = p \frac{6}{2(\frac{1}{16})} = 48p$ psi
 $\sigma_r = \sigma_z = 0$

Bolts : Force = $p \frac{\pi}{4}(D^2) = p \frac{\pi}{4}(6)^2 = 28.3p$ lb.

Stress $\sigma_z = \frac{28.3p}{3} \frac{1}{\frac{\pi}{4}(\frac{1}{4})^2} = 192p$ psi

FORCE-DEFORMATION:

Cylinder : $\epsilon_z = \frac{1}{E_B} [\sigma_z - \nu_B(\sigma_\theta + \sigma_r)] = \frac{-\nu_B \sigma_\theta}{E_B}$

Bolts : $\epsilon_z = \frac{\sigma_z}{E_S}$

Thus: $\delta_2 = (-\epsilon_z)_{\text{cylinder}} (L)_{\text{cyl.}} = -\left(\frac{-\nu_B}{E_B} 48p\right) 9$

and $\delta_1 = (\epsilon_z)_{\text{bolt}} (L)_{\text{bolt}} = \left(\frac{192p}{E_S}\right) 10$

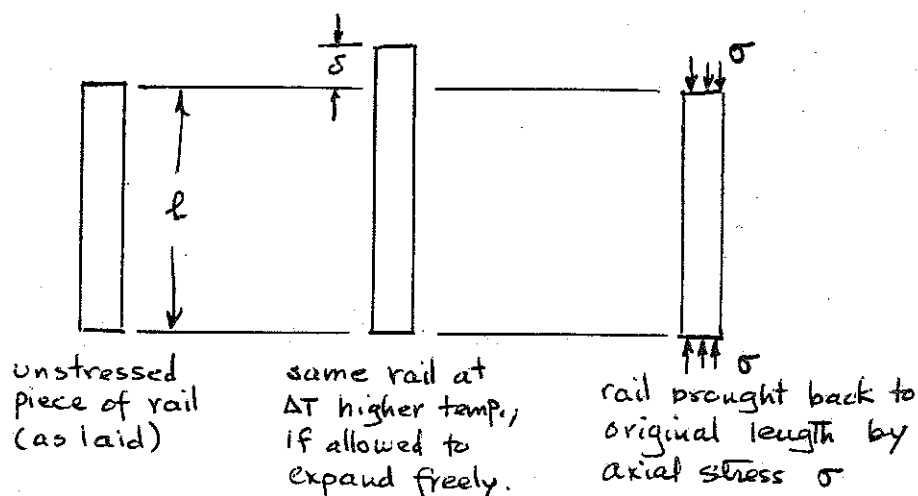
From (1) above : $\frac{(9)(48)\nu_B p}{E_B} + \frac{1920p}{E_S} = 0.025$

And $\nu_B = \nu_{\text{BRASS}} = 0.34$; $E_B = 15.2 \times 10^6$ psi ; $E_S = E_{\text{STEEL}} = 30 \times 10^6$ psi

Thus: $\frac{(9)(48)(0.34)(p)}{15.2 \times 10^6} + \frac{1920p}{30 \times 10^6} = 0.025$

Giving p = 338 psi

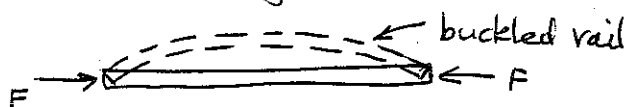
5.15 If we suppose that the ends of a piece of straight rail are not allowed to move, then thermal stresses will be set up as the rail heats and cools.



$$\delta = \alpha l \Delta T \quad \therefore \quad \underline{\sigma = E \frac{\delta}{l} = E \alpha (\Delta T)}$$

Thus we find that stresses of magnitude $E \alpha \Delta T$ are required to be exerted on the ends. But that is not all.

Under the action of the end forces the rail will tend to buckle, and the longer the rail the greater the tendency to buckle.



Lateral constraint is necessary to prevent buckling, and the spacing of lateral supports will depend on rail cross section and elastic constants.

The buckling problem does not arise when F is tensile (ΔT negative), unless the rail is curved, in which case it will tend to straighten out.

5.16 Stresses are caused by unequal thermal expansion. If the cladding thickness is small compared to the plate thickness, the stresses in the plate will be negligibly small.

SYMMETRY: shear stresses = 0 (except at ends)

GEOMETRY: $(\epsilon_x)_{\text{clad}} = (\epsilon_x)_{\text{plate}}$
 $(\epsilon_y)_{\text{clad}} = (\epsilon_y)_{\text{plate}}$ } - ①

STRESS-STRAIN-TEMP:

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] + \alpha \Delta T \text{ etc.}$$

EQUILIBRIUM: $\sigma_z = 0$

(Notation: Subscript p for plate; c for cladding)

Plate: all stresses ≈ 0

$$\therefore (\epsilon_x)_p = \alpha_p \Delta T = (\epsilon_y)_p$$

Cladding $(\epsilon_x)_c = \frac{1}{E_c} [(\sigma_x)_c - \nu_c(\sigma_y)_c] + \alpha_c \Delta T = \alpha_p \Delta T$
 $(\epsilon_y)_c = \frac{1}{E_c} [(\sigma_y)_c - \nu_c(\sigma_x)_c] + \alpha_c \Delta T = \alpha_p \Delta T$ } from ①

Solving: $(\sigma_x)_c = (\sigma_y)_c = \frac{E_c(\alpha_p - \alpha_c) \Delta T}{1 - \nu_c}$. These are two principal stresses. The third is $\sigma_z = 0$.

YIELD CRITERION: Use Tresca (max. shear stress)

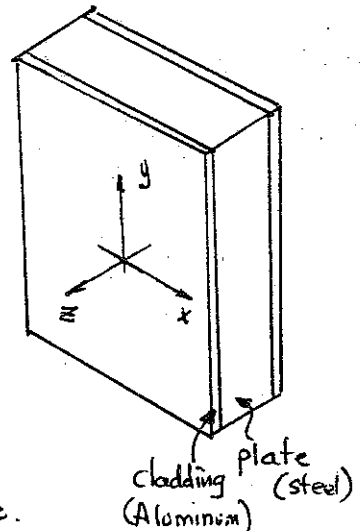
$$\tau_{\max} = \frac{1}{2} Y$$

Cladding: $\tau_{\max} = \frac{1}{2} (\sigma_{\max} - \sigma_{\min}) = \frac{1}{2} \left[\frac{E_c(\alpha_p - \alpha_c) \Delta T}{1 - \nu_c} - 0 \right]$

Properties: $\alpha_c = 12 \times 10^{-6} \text{ } ^\circ\text{F}^{-1}$ $\alpha_p = 6 \times 10^{-6} \text{ } ^\circ\text{F}^{-1}$
 $\nu_c = 0.33$; $E_c = 11 \times 10^6 \text{ psi}$ $Y_c = 5000 \text{ psi}$;

$$\therefore \frac{5000}{2} = \frac{11 \times 10^6 (12 - 6) \times 10^{-6} (\Delta T)}{2(1 - 0.33)} \quad \text{or} \quad \Delta T = 51^\circ\text{F} = 28.3^\circ\text{C}$$

[Note: Von Mises yield criterion gives the same answer]

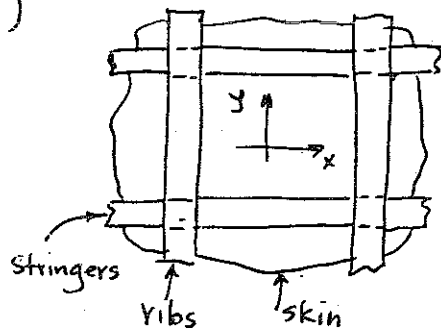


5.17 Heating the pulley or cooling the shaft will be both equally effective, although it may be easier to heat the pulley. Also, when the pulley is to be fixed on the shaft at some distance from the shaft ends it will be much more convenient to heat the pulley.

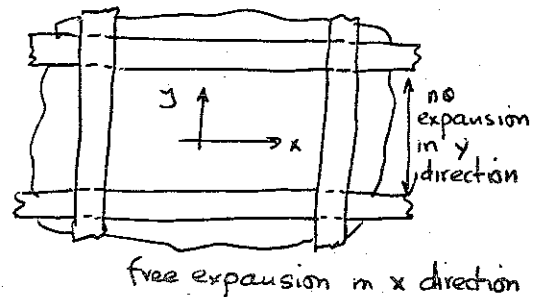
Since the pulley hole dia. is 0.050 mm less than the shaft dia., and a 0.025 clearance during fitting is required, the thermal expansion should be 0.075 mm, over a 25 mm length.

$$\therefore \epsilon = \frac{0.075}{25} = \alpha \Delta T \quad \alpha \Delta T = \frac{0.003}{11 \times 10^{-6}} = \underline{272^\circ\text{C}}$$

5.18 1)



Uniform temp



Skin hotter than ribs

$\Delta T = 50^\circ\text{F}$

Ribs prevent lateral strain.

GEOMETRY : $\epsilon_y = 0$

EQUILIBRIUM : $\sigma_x = \sigma_z = \text{shear stresses} = 0$

STRESS-STRAIN-TEMP : $\epsilon_y = \frac{1}{E}[\sigma_y - \nu(\sigma_x + \sigma_z)] + \alpha \Delta T = 0$ (from above)

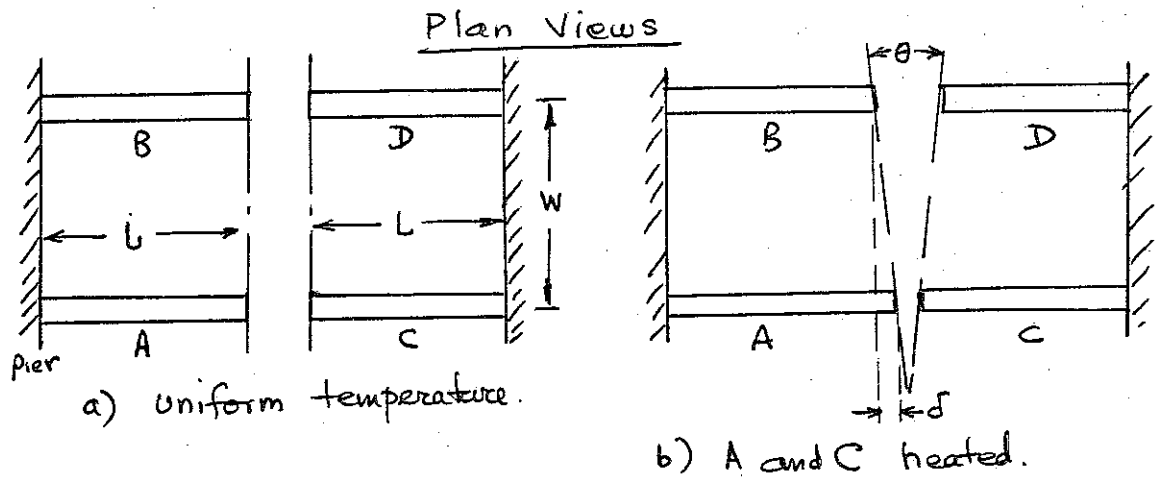
$$\therefore \sigma_y = -E\alpha \Delta T = -(11 \times 10^6)(12 \times 10^{-6})(50) = \underline{-6600 \text{ psi}}$$

$$\epsilon_y = \frac{1}{E}[\sigma_y - \nu(\sigma_x + \sigma_z)] + \alpha \Delta T = -\frac{(0.33)(-6600)}{11 \times 10^6} + (12 \times 10^{-6})(50) = \underline{800 \times 10^{-6}}$$

$$\epsilon_z = \frac{1}{E}[\sigma_z - \nu(\sigma_x + \sigma_y)] + \alpha \Delta T = \underline{800 \times 10^{-6}}$$

2) Ribs offer no restraint : $\text{stresses} = 0$; $\epsilon_x = \epsilon_y = \epsilon_z = \underline{600 \times 10^{-6}}$

5.19 We will represent each truss by just two members. The sun will heat the members on one side.



Expansion of A and C = $(\alpha \Delta T)L = \delta$.

The ends of the two spans will have an angular mismatch θ where $\theta \approx \frac{2\delta}{W} = 2(\alpha \Delta T)(\frac{L}{W})$

Estimate: $L = 200 \text{ ft}$; $W = 30 \text{ ft}$; $\Delta T = 50^\circ \text{F}$; and $\alpha = 6 \times 10^{-6} \text{ } ^\circ \text{F}^{-1}$ (steel)

$$\theta = \frac{2(6 \times 10^{-6})(50)(200)}{(30)} = 0.004 \text{ radian} = \underline{0.2 \text{ degrees}}$$

5.20 GEOMETRY: $\epsilon_z = 0$ in cylinder wall.

EQUILIBRIUM:

Free body: $F + R = p\pi r^2$

$$\therefore R = p\pi r^2 - F$$

$\therefore$ axial stress in cylinder

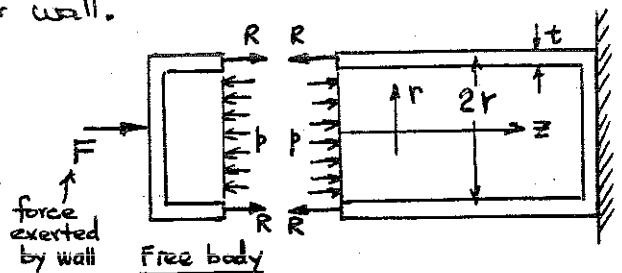
$$= \frac{R}{2\pi r t} = \frac{p\pi r^2 - F}{2\pi r t} = \sigma_z$$

Circumferential stress = $\sigma_\theta = \frac{pr}{t}$ and $\sigma_r \approx 0$

STRESS-STRAIN: $\epsilon_z = \frac{1}{E}[\sigma_z - \nu(\sigma_r + \sigma_\theta)] = 0$

$$\therefore \sigma_z = \nu(\sigma_r + \sigma_\theta) \quad \text{i.e.} \quad \frac{p\pi r^2 - F}{2\pi r t} = \nu \frac{pr}{t}$$

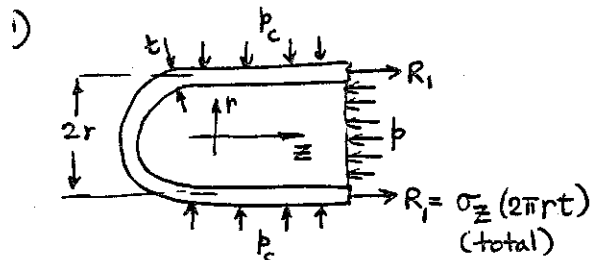
$$\therefore F = p\pi r^2(1 - 2\nu)$$



5.21 IDEALISATION: Zero friction between tank and walls of cavity.
 Let p_c = contact pressure between tank and cavity.

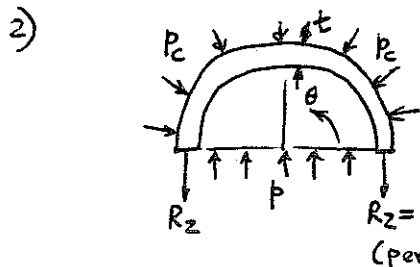
GEOMETRY: $\epsilon_\theta = 0$ in tank wall.

EQUILIBRIUM:



$$\sum F_z = 0 : p\pi r^2 = R_1 = \sigma_z(2\pi r t)$$

$$\therefore \sigma_z = \frac{pr}{2t}$$



$$\sum F_{\text{vertical}} = 0 :$$

$$2\sigma_\theta t = (p - p_c)(2r)$$

$$R_2 = \sigma_\theta(t) \quad \therefore \sigma_\theta = \frac{(p - p_c)r}{t}$$

And σ_r is between $-p$ and $-p_c$. Assuming that p_c is of order p we will neglect σ_r . ($t/r \ll 1$)

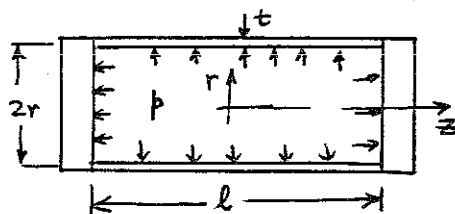
STRESS-STRAIN: $\epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_r + \sigma_z)] = 0$

$$\therefore \sigma_\theta = \nu(\sigma_r + \sigma_z) = \nu(\sigma_z) = \frac{\nu pr}{2t}$$

[and $p_c = p - \frac{t}{r}\sigma_\theta = p(1 - \frac{\nu}{2}) \therefore$ assumption above, ok]

5.22 GEOMETRY: Rigid ends prevent tangential strain (ϵ_θ). But if (l/r) is large then this end effect may be neglected.

EQUILIBRIUM: $\sigma_z = \frac{pr}{2t}$; $\sigma_\theta = \frac{pr}{t}$; $\sigma_r \approx 0$



STRESS-STRAIN: $\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_\theta + \sigma_r)] = \frac{pr}{2Et} (1 - 2\nu)$

$$\epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_r + \sigma_z)] = \frac{pr}{2Et} (2 - \nu)$$

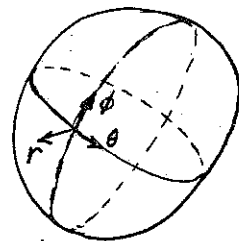
$$\left. \begin{aligned} \delta_D = (D) \epsilon_\theta = 2r \epsilon_\theta \\ \delta_L = l \epsilon_z \end{aligned} \right\} \therefore \frac{\delta_L}{\delta_D} = \frac{l}{2r} \cdot \frac{(1 - 2\nu)}{(2 - \nu)}$$

5.23 EQUILIBRIUM: Consider the equilibrium of a hemisphere. This gives
 $\sigma_\theta = \sigma_\phi = \frac{pr}{2t}$; and $\sigma_r \approx 0$

STRESS-STRAIN:

$$\epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_\phi + \sigma_r)] = \frac{pr}{2Et} (1-\nu)$$

$$\text{and } \delta_D = (D)\epsilon_\theta = 2r\epsilon_\theta = \frac{pr^2}{Et} (1-\nu)$$



internal pressure = p
 radius = r

$$\text{From prob 5.22: } (\delta_D)_{\text{cylinder}} = \frac{pr^2}{Et} (2-\nu)$$

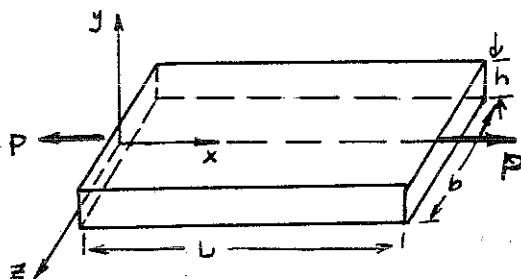
$$\therefore \frac{(\delta_D)_{\text{cylinder}}}{(\delta_D)_{\text{sphere}}} = \left(\frac{pr^2}{Et} (2-\nu) \right) \left(\frac{Et}{pr^2(1-\nu)} \right) = \frac{2-\nu}{1-\nu}$$

5.24 OVERALL EQUILIBRIUM: the following stresses on the bounding surfaces satisfy overall equilibrium:

$$\sigma_x = \frac{P}{bh}; \quad \sigma_y = \sigma_z = 0$$

$$\tau_{xy} = \tau_{yz} = \tau_{zx} = 0$$

We will start by assuming that these are also the stresses within the body.



$$\begin{aligned} \text{STRESS-STRAIN: } \epsilon_x &= \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)] = \frac{P}{Ebh} \\ \epsilon_y &= -\frac{\nu P}{Ebh} \\ \epsilon_z &= -\frac{\nu P}{Ebh} \end{aligned} \quad \left. \begin{array}{l} \text{shear} \\ \text{strains} = 0 \end{array} \right\} \text{①}$$

$$\text{Thus: } \frac{\partial u}{\partial x} = \frac{P}{Ebh}; \quad \frac{\partial v}{\partial y} = -\frac{\nu P}{Ebh} = \frac{\partial w}{\partial z}; \quad \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \text{ etc} = 0$$

Integrating, with $u=v=w=0$ at the origin, we obtain:

$$u = \frac{Px}{Ebh}; \quad v = -\frac{\nu Py}{Ebh}; \quad w = -\frac{\nu Pz}{Ebh} \quad \text{PRINCIPLES OF ELASTICITY:}$$

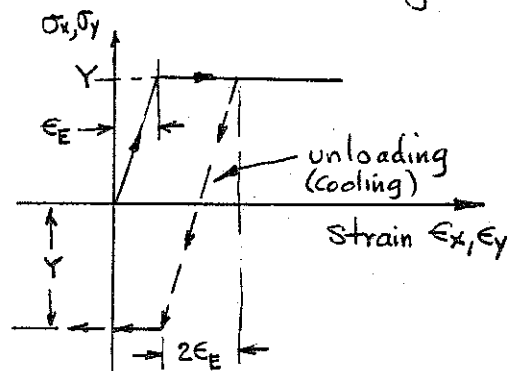
- 1) EQUILIBRIUM: all derivatives of stress = 0 $\therefore$ equil. eqns. satisfied.
- 2) GEOMETRIC COMPATIBILITY: displacements are single valued, continuous
- 3) STRESS-STRAIN LAWS: satisfied by ① above.

5.25 When the plate is heated to 50°C above the temperature which initiates yielding (i.e. about 80°C above ambient - see prob 5.16) the cladding will be fully plastic, with $\sigma_x = \sigma_y = Y$.

As the temperature is lowered again, the cladding unloads elastically. It will become plastic again if the strain during cooling exceeds $2\epsilon_E$ (see figure)

Now approx. 28°C temperature change causes a strain of ϵ_E (prob 5.16). When the plate is cooled from 80°C above ambient the strain will be about $3\epsilon_E$ which is $> 2\epsilon_E$.

The cladding will therefore be plastic at ambient temperature, with $\sigma_x = \sigma_y = -Y$



5.26 Unequal thermal expansion will 1) cause unequal changes in the diameters of the two hoops and as contact pressure "p" at the interface; 2) cause unequal lateral expansion. If there is high friction between the hoops or if they are bonded, this lateral expansion will introduce axial stresses and shear stresses. To simplify the analysis we will assume that the hoops are free to slide on each other, so that the lateral expansions are unrestrained.

Since (t/r) is small for both hoops we will use the same mean radius for both hoops

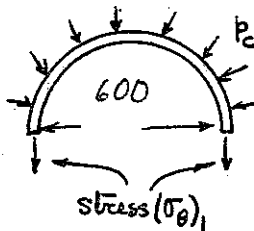
The mean radius increases by δ_1 after heating.

GEOMETRY: tangential strain ϵ_θ in both hoops = $\frac{\delta_1}{300 \text{ mm}}$

SYMMETRY: $\tau_{r\theta} = 0$ at interface

EQUILIBRIUM: $\sigma_z = 0$; Assume $\sigma_r = 0$

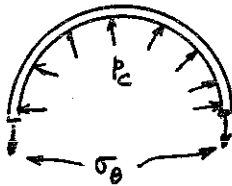
a) Inner hoop: $\sum F_v = 0$



$$p_c(600)(60) = \sigma_\theta 3 \times 60 \times 2$$

$$\sigma_{\theta 1} = 100 p_c$$

b) Outer hoop



$$\sum F_{\text{vertical}} = 0 :$$

$$(\sigma_\theta)_2 = 50 p_c$$

STRESS-STRAIN: $\epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_z + \sigma_r)] + \alpha \Delta T = \frac{\sigma_\theta}{E} + \alpha \Delta T$

Thus: inner hoop: $(\epsilon_\theta)_1 = \frac{-100 p_c}{E_1} + \alpha_1 \Delta T = \delta_1 /$
 outer hoop: $(\epsilon_\theta)_2 = \frac{50 p_c}{E_2} + \alpha_2 \Delta T = \delta_1 /$ } from geometry.

$$\therefore \frac{-100 p_c}{E_1} + \alpha_1 \Delta T = \frac{50 p_c}{E_2} + \alpha_2 \Delta T$$

Properties: ①: (brass) - $E_1 = 105 \text{ GN/m}^2$; $\alpha_1 = 20.3 \times 10^{-6} / ^\circ\text{C}$
 ②: (steel) - $E_2 = 210 \text{ GN/m}^2$; $\alpha_2 = 10.8 \times 10^{-6} / ^\circ\text{C}$

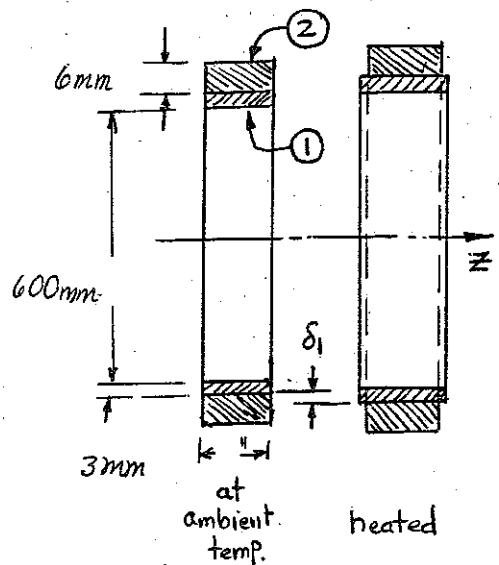
$$\frac{p_c}{E_1} [100 + 50 \frac{E_1}{E_2}] = \Delta T (\alpha_1 - \alpha_2) = 85 (9.5) \times 10^{-6}$$

$$p_c = 0.678 \text{ MN/m}^2$$

BRASS Hoop $\sigma_\theta = -100 p_c = -67.8 \text{ MN/m}^2$

STEEL Hoop $\sigma_\theta = 50 p_c = 33.9 \text{ MN/m}^2$

} ELASTIC (ok)



5.27 Find the principal stresses, using Mohr's circle in the σ - τ plane:

Principal Stresses:

$$\sigma_1 = 207 \text{ MN/m}^2$$

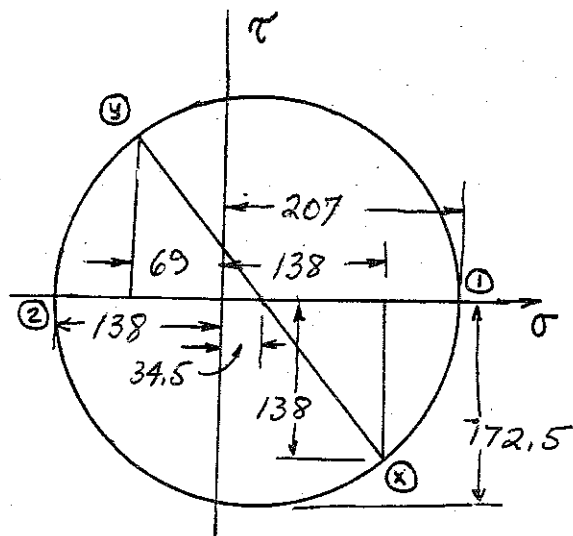
$$\sigma_2 = -138 \text{ MN/m}^2$$

$$\sigma_3 = \sigma_z = 0$$

a) MISES YIELD CRITERION

$$\bar{\sigma} = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]} = Y$$

$$\bar{\sigma} = 301 \text{ MN/m}^2 < Y$$



∴ will not yield.

b) MAX. SHEAR STRESS CRITERION: $|\sigma_{\max} - \sigma_{\min}| = Y$

$$\sigma_{\max} = \sigma_1 = 207 \text{ MN/m}^2; \sigma_{\min} = \sigma_2 = -138 \text{ MN/m}^2$$

$$\therefore |\sigma_{\max} - \sigma_{\min}| = 345 \text{ MN/m}^2 \text{ which is greater than } Y$$

∴ will yield.

5.28 p = internal pressure.

EQUILIBRIUM: $\sigma_\theta = \frac{PR}{t}$; $\sigma_z = \frac{PR}{2t} = \frac{1}{2}\sigma_\theta$
 $\sigma_r \approx 0$.

YIELD CRITERION: (MAX. SHEAR STRESS)

$$|\sigma_{\max} - \sigma_{\min}| = Y; \sigma_{\max} = \sigma_\theta; \sigma_{\min} = 0.$$

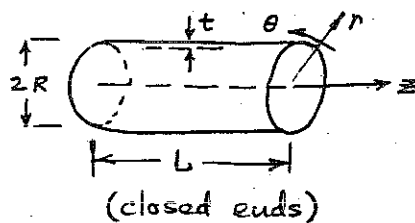
and $Y = 36000 \text{ psi}$ for 1020HR steel. ∴ $\sigma_\theta = Y = 36000 \text{ psi}$

and $\sigma_z = \frac{1}{2}\sigma_\theta = 18000 \text{ psi}$

STRESS STRAIN (ELASTIC) $\epsilon_\theta = \frac{1}{E} [\sigma_\theta - \nu(\sigma_r + \sigma_z)] = \frac{36000 - (0.28)(18000)}{30 \times 10^6}$

$$= 1.0 \times 10^{-3}; \epsilon_z = \frac{18000 - (0.28)(36000)}{30 \times 10^6} = 2.65 \times 10^{-4}$$

∴ $\frac{\delta(\text{LENGTH})}{\text{LENGTH}} = \epsilon_z = 0.0265\%$; $\frac{\delta(\text{CIRCUMFERENCE})}{\text{CIRCUMFERENCE}} = \epsilon_\theta = 0.1\%$



5.29 EQUILIBRIUM $\sigma_z = \frac{PR}{2z}$; $\sigma_\theta = \frac{PR}{z}$; $\sigma_r \approx 0$

STRESS STRAIN (PLASTIC)

EQNS 5.19 :

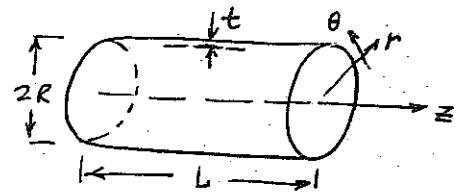
$$d\epsilon_z^p = \frac{d\bar{\epsilon}}{\bar{\sigma}} \left[\sigma_z - \frac{1}{2}\sigma_\theta \right] = 0$$

$$d\epsilon_\theta^p = \frac{d\bar{\epsilon}}{\bar{\sigma}} \left[\sigma_\theta - \frac{1}{2}\sigma_z \right] = \left(\frac{3}{4} \right) \left(\frac{PR}{z} \right) \frac{d\bar{\epsilon}}{\bar{\sigma}}$$

Thus, $\epsilon_z^p = \int d\epsilon_z^p = 0$; $\epsilon_\theta^p = \int d\epsilon_\theta^p = \int \left(\frac{3}{4} \right) \left(\frac{PR}{z} \right) \frac{d\bar{\epsilon}}{\bar{\sigma}} \neq 0 = \epsilon_0$ say

Neglecting elastic strains, $\epsilon_z = \epsilon_z^p$; $\epsilon_\theta = \epsilon_\theta^p$

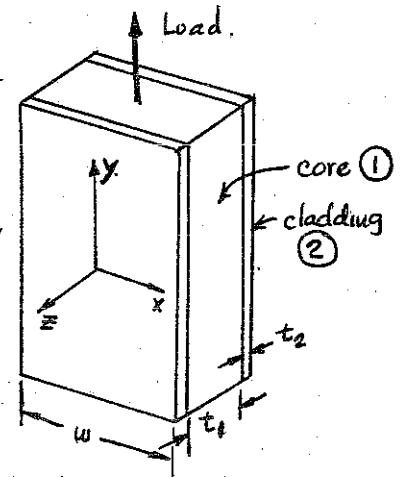
$$\therefore \frac{\text{axial strain}}{\text{tangential strain}} = \frac{\epsilon_z}{\epsilon_\theta} = \frac{0}{\epsilon_0} = 0$$



closed ends ; internal pressure p.

5.30 In an actual test the width w of the specimen would be considerably larger than the thickness since the thickness is so small (0.051 in). While an analysis for this case is possible, it is fairly complex mainly because the σ_x stresses cannot be neglected.

It is a case of generalised plane strain. For simplicity we will make the unrealistic assumption that w is small compared to t_1 .



Then, since the stresses σ_x , τ_{xy} and τ_{xz} are zero along the edges we may take them to be zero throughout.

SYMMETRY : $\tau_{yz} = 0$.

EQUILIBRIUM : $\sigma_z = 0$

Thus the only stress is σ_y .

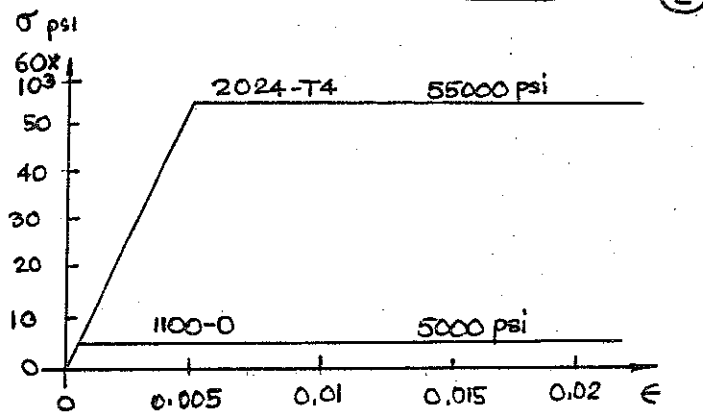
After unloading, the net y direction force = 0.

$$\sum F_y = 0 : (\sigma_y)_1 t_1 + 2(\sigma_y)_2 t_2 = 0 \quad \text{i.e. } (\sigma_y)_1 = -2 \left(\frac{t_2}{t_1} \right) (\sigma_y)_2 \quad \text{--- ①}$$

GEOMETRY : $(\epsilon_y)_1 = (\epsilon_y)_2$

STRESS-STRAIN DATA

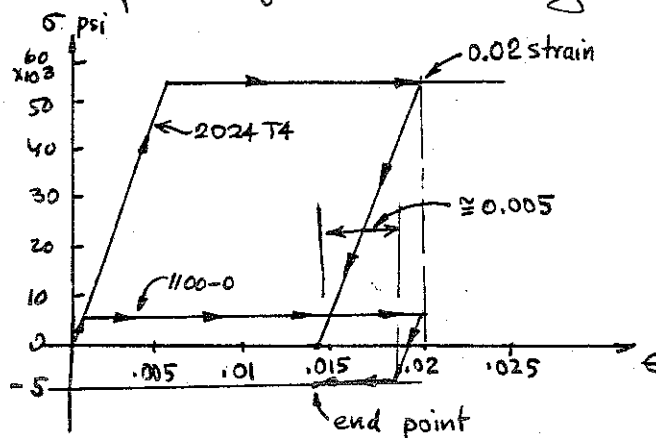
From Fig 5.5b, The elastic-perfectly plastic models shown alongside can be constructed.



- a) Here we neglect the second term in eqn ① above, which becomes $(\sigma_y)_1 = 0$. This determines the end point of stress-strain history for the core. Using ② we can then find the end point for the cladding.

This is graphically carried out in the figure alongside.

Thus: Residual stresses are
CORE $\cong 0$
CLADDING : -5000 psi



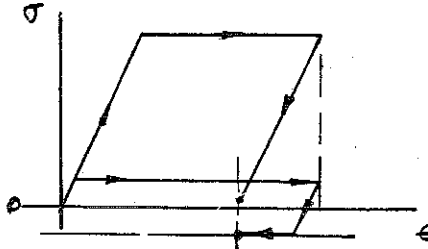
- b) For $t_1 = 0.041''$ and $t_2 = 0.005''$, eqn ① becomes:
 $(\sigma_y)_1 = -0.244(\sigma_y)_2$. In this case, both end points will be shifted to the right on the strain axis. Assume that this shift is less than about 0.005. Then $(\sigma_y)_2 = -5000$

and $(\sigma_y)_1 = -0.244(-5000) = 1220 \text{ psi}$

[shift in end point $\cong \frac{1220}{E} < 0.005$]

Residual stresses:

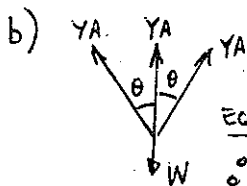
CORE 1220 psi
CLADDING -5000 psi



5.31 a) GEOMETRY little change during elastic deformation $\therefore \theta \sim \text{constant}$.

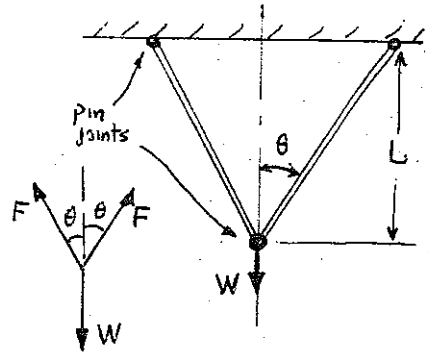
EQUILIBRIUM : $2F \cos \theta = W \therefore F = \frac{W}{2 \cos \theta}$

YIELDING : $F = Y A \therefore W = 2 Y A \cos \theta$



EQUILIBRIUM : $W = Y A [1 + 2 \cos \theta]$

$\therefore \text{increase} = Y A$



c) GEOMETRY : $\delta_1 = \delta_3 = \delta \cos \theta$

$\delta_2 = \delta$

Strains in the three rods are :

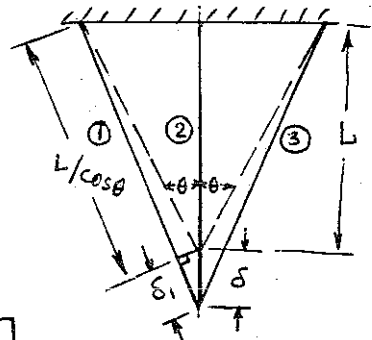
$\epsilon_1 = \frac{\delta_1}{L/\cos \theta} = \frac{\delta \cos \theta}{L/\cos \theta} = \frac{\delta}{L} \cos^2 \theta$; $\epsilon_2 = \frac{\delta}{L}$

STRESS-STRAIN (ELASTIC) $F = A(E\epsilon)$

$F_1 = F_3 = E A \frac{\delta}{L} \cos^2 \theta$; $F_2 = E A \frac{\delta}{L}$

EQUILIBRIUM : $W = 2 F_1 \cos \theta + F_2 = E A \frac{\delta}{L} [1 + 2 \cos^3 \theta]$

ie $\delta = \frac{W L / E A}{1 + 2 \cos^3 \theta}$



d) From (c), elastic stresses due to a load W are :

$\sigma_1 = \sigma_3 = \frac{F_1}{A} = E \frac{\delta}{L} \cos^2 \theta = \frac{(W/A) \cos^2 \theta}{1 + 2 \cos^3 \theta}$; $\sigma_2 = \frac{(W/A)}{1 + 2 \cos^3 \theta}$

Plastic load from (b) = $W_p = Y A [1 + 2 \cos \theta]$. Assume elastic unloading, find stresses due to $-W_p$ and add these to the yield stresses. Thus for $W = -W_p$:

$\sigma_1 = \sigma_3 = - \frac{Y(1 + 2 \cos \theta) \cos^2 \theta}{1 + 2 \cos^3 \theta}$; $\sigma_2 = - \frac{Y(1 + 2 \cos \theta)}{1 + 2 \cos^3 \theta}$

Residual stresses : Rods ① and ③ : $Y - \frac{Y(1 + 2 \cos \theta) \cos^2 \theta}{1 + 2 \cos^3 \theta} = Y \left(\frac{1 - \cos^2 \theta}{1 + 2 \cos^3 \theta} \right)$

Rod ② : $Y - \frac{Y(1 + 2 \cos \theta)}{1 + 2 \cos^3 \theta} = - \frac{2 Y \sin^2 \theta \cos \theta}{1 + 2 \cos^3 \theta}$ [Both are $< Y$ $\therefore$ elastic unloading OK]

Check : $\theta = 0$: all rods in uniaxial tension. Hence residual stress should be zero. This checks.

$\theta = \pi/2$: ① and ③ ineffective. ② in simple tension. $\therefore$ Residual stress = 0. Checks.

5.32 IDEALISATION pin joints everywhere;
 θ small.

GEOMETRY: $\delta_1 = c\theta$; $\delta_2 = 3c\theta$

$$\therefore \epsilon_1 = \frac{c\theta}{L}; \quad \epsilon_2 = \frac{3c\theta}{L}$$

EQUILIBRIUM: $\sum M_{\odot} = 0$

$$F_1(c) + F_2(3c) = P(2c)$$

$$\text{i.e. } F_1 + 3F_2 = 2P$$

STRESS-STRAIN (ELASTIC)

$$F = \sigma A = E \epsilon A$$

$$F_1 = \frac{EAc\theta}{L}; \quad F_2 = \frac{3EAc\theta}{L}$$

$$\therefore \left(\frac{EAc\theta}{L} \right) + 3 \left(\frac{3EAc\theta}{L} \right) = 2P$$

$$\text{or } \frac{EAc\theta}{L} = \frac{P}{5} \quad \text{i.e. } F_1 = \frac{P}{5}; \quad F_2 = \frac{3P}{5}$$

$$\text{and } \sigma_1 = \frac{P}{5A}; \quad \sigma_2 = \frac{3P}{5A}$$

Both rods are plastic when rod ① is plastic.

$$F_1 = F_2 = YA \quad \text{and } P = \frac{1}{2}(F_1 + 3F_2) = 2YA$$

Residual stress = plastic stress + change in stress on unloading. Assume unloading is elastic (i.e. causes no reverse plastic flow). Elastic stresses due to a load $-2YA$ are:

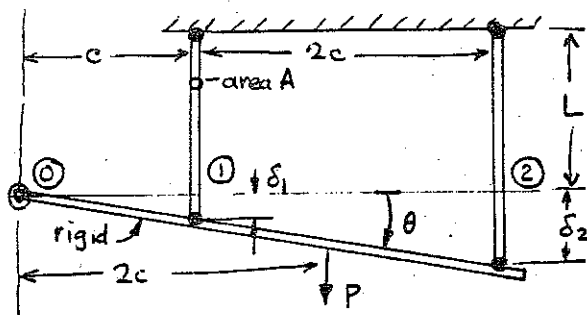
$$\sigma_1 = -\frac{2}{5}Y; \quad \sigma_2 = -\frac{6}{5}Y$$

Residual stresses: Rod ①: $Y - \frac{2}{5}Y = \frac{3}{5}Y$
 Rod ②: $Y - \frac{6}{5}Y = -\frac{1}{5}Y$ } These are less than Y (absolute magnitude) $\therefore$ elastic unloading OK.

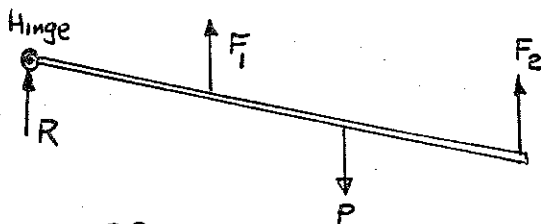
At the end of the loading cycle, when rod ① has just become plastic, $\sigma_1 = E\left(\frac{c\theta}{L}\right) = Y \therefore \theta = \frac{YL}{Ec}$

During unloading (elastic) the angle change is given by $\frac{EAc\theta}{L} = \frac{P}{5}$ where $P = -2YA \therefore \theta = -\frac{2}{5} \frac{YL}{Ec}$

$$\therefore \text{Residual angle} = \frac{YL}{Ec} - \frac{2}{5} \frac{YL}{Ec} = \frac{3}{5} \left(\frac{YL}{Ec} \right)$$



Free body:



5.33

1020 CR max. uniform strain ≈ 0.01
 1020 HR max. uniform strain ≈ 0.20

$$\therefore \frac{(\Delta L)_{CR}}{(\Delta L)_{HR}} \approx \frac{L(0.01)}{L(0.20)} = \underline{0.05}$$

True strains at fracture : 1020 CR : $\epsilon_f = 0.70$
 1020 HR : $\epsilon_f = 0.90$

$$\therefore \frac{(\epsilon_f)_{CR}}{(\epsilon_f)_{HR}} = \frac{0.7}{0.9} = \underline{0.78}$$

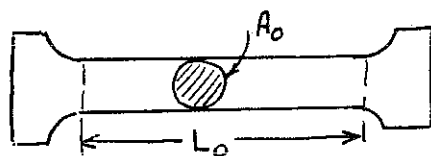
Let us measure the relative ductility, D_R , of the two materials by the ratio of the average strains to fracture.

For very long specimens, average strain $\approx$ uniform strain
 and $D_R = \frac{1020CR \text{ ductility}}{1020HR \text{ ductility}} = \underline{0.05}$

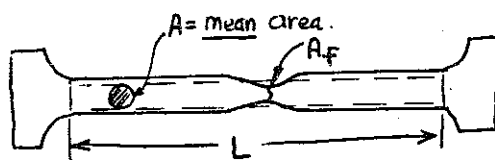
For very short specimens, almost the entire specimen will neck and average strain $\approx$ fracture strain, and
 $D_R = 0.78$

The change is striking.

5.34



before strain



fracture

We will neglect elastic strains.

$$\% \text{ Elong} = PE = 100 \left(\frac{L - L_0}{L_0} \right) ; \% \text{ Reduction in Area} = RA = 100 \left(\frac{A_0 - A_f}{A_0} \right)$$

Volume Constancy (plastic flow) $A_0 L_0 = AL$

Also, $A_f \leq A$ (see figure) $A_f = A$ for no necking ; $A_f < A$ for necking.

$$\text{Thus } \underline{A_0 L_0 \geq A_f L} ; L = L_0 \left[1 + \frac{PE}{100} \right] ; A_f = A_0 \left[1 - \frac{RA}{100} \right]$$

$$\therefore A_0 L_0 \geq A_0 \left[1 - \frac{RA}{100} \right] L_0 \left[1 + \frac{PE}{100} \right]$$

i.e. $\left[1 - \frac{RA}{100}\right] \left[1 + \frac{PE}{100}\right] \leq 1$ where the equality holds when there is no necking. Hence if the quantity $\left(1 - \frac{RA}{100}\right) \left(1 + \frac{PE}{100}\right) < 1$ then there was necking before fracture. If $= 1$ then fracture occurred without necking.

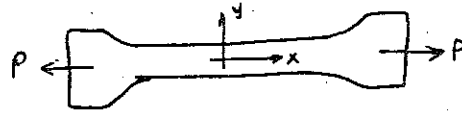
5.35 All stresses except σ_x are zero.

From 5.14, $\bar{\sigma} = \sigma_x$

$$\text{5.19: } d\epsilon_x^p = \frac{d\bar{\epsilon}^p}{\bar{\sigma}} [\sigma_x - 0] = \frac{d\bar{\epsilon}^p \sigma_x}{\bar{\sigma}}$$

$$= d\bar{\epsilon}^p$$

$$\text{Thus: } \bar{\epsilon}^p = \int d\bar{\epsilon}^p = \int d\epsilon_x^p = \epsilon_x^p$$



5.36 Eg. (i) of Section (5-7): $u = Ar + \frac{B}{r}$ B.C. $\sigma_r = -p_i$ @ $r = r_i$
 $\epsilon_r = \frac{\partial u}{\partial r}$; $\epsilon_\theta = \frac{u}{r}$ $\sigma_r = -p_o$ @ $r = r_o$

$$\therefore \epsilon_r = A - \frac{B}{r^2}; \quad \epsilon_\theta = A + \frac{B}{r^2}$$

Relationship between σ_r and $\epsilon_r, \epsilon_\theta$

$$\sigma_r = \frac{E}{(1-\nu^2)} [\epsilon_r + \nu \epsilon_\theta] = \frac{E}{(1-\nu^2)} \left[\left(A - \frac{B}{r^2}\right) + \nu \left(A + \frac{B}{r^2}\right) \right] = \frac{E}{(1-\nu^2)} \left[A(1+\nu) - \frac{B}{r^2}(1-\nu) \right]$$

Using B.C. @ $r = r_i$

$$(\sigma_r)_{r=r_i} = -p_i = \frac{E}{(1-\nu^2)} \left[A(1+\nu) - \frac{B}{r_i^2}(1-\nu) \right]; \quad \text{@ } r = r_o$$

$$-p_o = \frac{E}{(1-\nu^2)} \left[A(1+\nu) - \frac{B}{r_o^2}(1-\nu) \right]$$

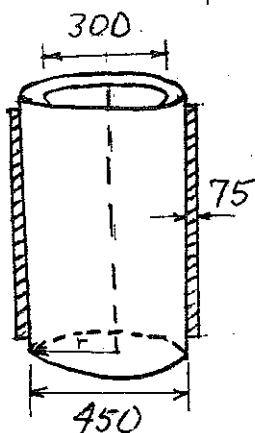
$$\rightarrow \frac{E}{(1-\nu^2)} \begin{bmatrix} (1+\nu) & -\frac{(1-\nu)}{r_i^2} \\ (1+\nu) & -\frac{(1-\nu)}{r_o^2} \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = - \begin{bmatrix} p_i \\ p_o \end{bmatrix}; \text{ solving for A and B.}$$

$$A = \frac{1-\nu}{E} \cdot \frac{p_o r_o^2 - p_i r_i^2}{r_i^2 - r_o^2}$$

$$B = \frac{(1+\nu)}{E} \cdot \frac{(p_o - p_i) r_i^2 r_o^2}{r_i^2 - r_o^2}$$

$$\therefore u(r) = \frac{1}{E \left[1 - \left(\frac{r}{r_o}\right)^2 \right]} \left\{ (1-\nu) \left[p_i \left(\frac{r_i}{r_o}\right)^2 - p_o \right] r + (1+\nu) \frac{r_i^2}{r} (p_i - p_o) \right\}$$

(5-37)



Assumptions: (i) Shaft diameter doesn't change appreciably
(ii) Sleeve wall thickness is approx. constant

From Problem (5-36)

$$u(r) = \frac{1}{E[1 - (\frac{r_i}{r_o})^2]} \left\{ (1-\nu) \left[p_i \left(\frac{r_i}{r} \right)^2 - p_o \right] r + (1+\nu) \frac{r_i^2}{r} (p_i - p_o) \right\}$$

For the sleeve: $r_i = 225$, $r_o = 300$, $p_o = 0$, $\frac{r_i}{r_o} = \frac{9}{12}$

$$u(225) = 0.625 \text{ mm} = \frac{p_i}{E[1 - (\frac{9}{12})^2]} \left\{ (1-\nu) \left(\frac{9}{12} \right)^2 (225) + (1+\nu) \frac{225^2}{225} \right\}$$

$$E = 205 \text{ GN/m}^2 ; \nu = 0.3$$

$$p_i = 147 \text{ MN/m}^2$$

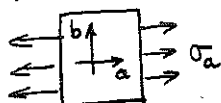
FOR THE SHAFT $r_i = 150 \text{ mm}$; $r_o = 225$; $p_o = 147 \text{ MN/m}^2$, $p_i = 0$

FROM Eq (5.9); SEE FIG 5.19

$$\frac{\sigma_r}{p_o} = - \frac{[(\frac{r_o}{r_i})^2 - (\frac{r_o}{r})^2]}{(\frac{r_o}{r_i})^2 - 1} ; \frac{\sigma_\theta}{p_o} = \frac{[(\frac{r_o}{r_i})^2 + (\frac{r_o}{r})^2]}{(\frac{r_o}{r_i})^2 - 1}$$

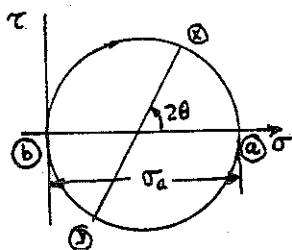
State of stress:

5.38



other stress = 0

Mohr's circle in the a-b plane:

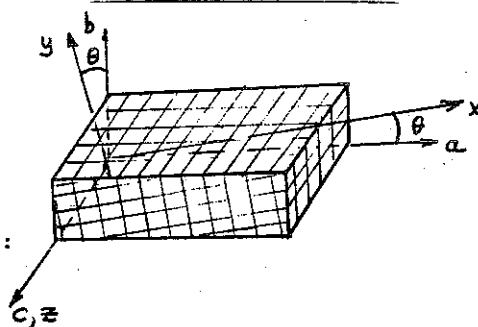


$$\sigma_x = \frac{\sigma_a}{2} (1 + \cos 2\theta)$$

$$\sigma_y = \frac{\sigma_a}{2} (1 - \cos 2\theta)$$

$$\tau_{xy} = -\frac{\sigma_a}{2} \sin 2\theta$$

other stresses = 0



STRESS-STRAIN:

$$\epsilon_x = S_{11}\sigma_x + S_{12}(\sigma_y + \sigma_z) = \frac{\sigma_a}{2} [(S_{11} + S_{12}) + (S_{11} - S_{12}) \cos 2\theta]$$

$$\epsilon_y = S_{11}\sigma_y + S_{12}(\sigma_x + \sigma_z) = \frac{\sigma_a}{2} [(S_{11} + S_{12}) - (S_{11} - S_{12}) \cos 2\theta]$$

$$\tau_{xy} = S_{44}\tau_{xy} = -\frac{\sigma_a}{2} [S_{44} \sin 2\theta]$$

Mohr's circle to transform these strains back to (a-b-c) coordinates.

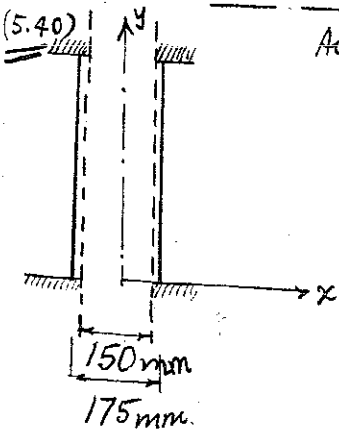
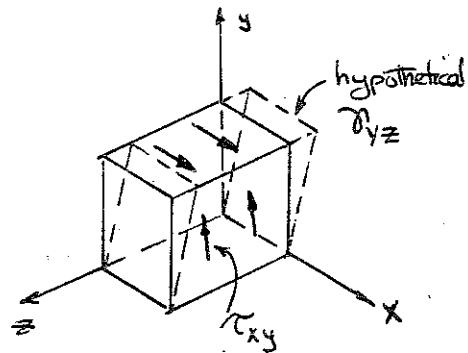
$$\begin{aligned}\text{Thus } \tau_{ab} &= (\epsilon_x - \epsilon_y) \sin 2\theta + \tau_{xy} \cos 2\theta \\ &= \frac{\sigma_a}{2} \left[2(S_{11} - S_{12}) \cos 2\theta \right] \sin 2\theta - \frac{\sigma_a}{2} [S_{44} \sin 2\theta] \cos 2\theta \\ \text{or } \left(\frac{\tau_{ab}}{\sigma_a} \right) &= \left[S_{11} - S_{12} - \frac{1}{2} S_{44} \right] \sin 2\theta \cos 2\theta\end{aligned}$$

A more meaningful measure of the shear strain may be the ratio $\left(\frac{\tau_{ab}}{\epsilon_a} \right)$. This works out to be:

$$\frac{\tau_{ab}}{\epsilon_a} = \frac{[S_{11} - S_{12} - \frac{1}{2} S_{44}] \sin 2\theta \cos 2\theta}{(S_{11} + S_{12}) + (S_{11} - S_{12}) \cos^2 2\theta + \frac{1}{2} S_{44} \sin^2 2\theta}$$

5.39

Let τ_{xy} produce τ_{yz}
(x, y, z are structural axes).
Rotate 180° about z axis.
 τ_{yz} is reversed but not τ_{xy} .
The x, y, z axes are still structural axes and the stress-strain behaviour should therefore be unchanged by the rotation.
Therefore if the relationship between τ and σ is linear, so that $\tau_{yz} = C \tau_{xy}$, then $C = 0$ and there are no shear strains. Similar argument for τ_{zx} .



Away from ends, assume stress distribution only in y -direction, i.e. $\sigma_x = \sigma_z = 0$

$$\begin{aligned}\therefore \epsilon_x &= -\frac{\nu \sigma_y}{E} + \alpha(T - T_0) \\ \epsilon_y &= \frac{\sigma_y}{E} + \alpha(T - T_0) \quad ; \quad \tau_{xy} = \tau_{yz} = \tau_{xz} = 0 \\ \epsilon_z &= -\frac{\nu \sigma_y}{E} + \alpha(T - T_0)\end{aligned}$$

Because of constraint by wall $\epsilon_y = 0 \rightarrow \sigma_y = -\alpha E \Delta T$
 $E = 205 \text{ GN/m}^2$; $\alpha = 12 \times 10^{-6} / ^\circ\text{C}$; $\Delta T = -35^\circ\text{C}$

$$\therefore \sigma_y = 86 \text{ MN/m}^2$$

$$\begin{aligned}\epsilon_x &= \alpha \Delta T (1 + \nu) \\ \epsilon_y &= \alpha \Delta T (1 + \nu)\end{aligned}$$

Note that all strains are constants

(5-41)

One dimensional stress state.

From turning the nut, strain in bolt: $\epsilon_{XB} = \frac{\sigma_{XB}}{E_B}$ For sleeve: change of length $\Delta L = -(\text{turn}) \times (\text{lead}) + \frac{\sigma_{XB} L}{E_B}$

$$\epsilon_{XS} = \frac{\Delta L}{L} = -\left(\frac{1}{4}\right)\left(\frac{1}{16}\right)\left(\frac{1}{L}\right) + \frac{\sigma_{XB}}{E_B}$$

$$\sigma_{XS} = \epsilon_{XS} E_S = \left\{ -\left(\frac{1}{4}\right)\left(\frac{1}{16}\right)\left(\frac{1}{L}\right) + \frac{\sigma_{XB}}{E_B} \right\} E_S$$

For $E_S = 10 \times 10^6 \text{ lb/in}^2$; $E_B = 30 \times 10^6 \text{ lb/in}^2$

$$\sigma_{XS} = \left\{ -3.8 \times 10^4 + \frac{\sigma_{XB}}{3} \right\} \text{ lb/in}^2 \quad \text{..... (1)}$$

From equilibrium, $\sigma_{XS} A_S + \sigma_{XB} A_B = 0$

$$\therefore \sigma_{XS} \frac{\pi}{4} \left[\left(\frac{3}{4}\right)^2 - \left(\frac{1}{2}\right)^2 \right] + \sigma_{XB} \frac{\pi}{4} \left(\frac{1}{2}\right)^2 = 0 \rightarrow 5\sigma_{XS} + 4\sigma_{XB} = 0$$

together with eq. (1),

$$5\left(-3.8 \times 10^4 + \frac{\sigma_{XB}}{3}\right) + 4\sigma_{XB} = 0 \rightarrow \sigma_{XB} = 3.36 \times 10^4 \text{ lb/in}^2$$

putting σ_{XB} into eq. (1),

$$\sigma_{XS} = \left(-3.8 \times 10^4 + \frac{3.36 \times 10^4}{3}\right) = -2.7 \times 10^4 \text{ lb/in}^2$$

We must also calculate stresses due to change in temperature

$$\epsilon_{XBT} = \frac{\sigma_{XBT}}{E_B} + \alpha_B \Delta T = \frac{\sigma_{XBT}}{30 \times 10^6} + 2.6 \times 10^{-4}$$

$$\epsilon_{XST} = \frac{\sigma_{XST}}{10 \times 10^6} + \alpha_S \Delta T = \frac{\sigma_{XST}}{10 \times 10^6} + 4.8 \times 10^{-4}$$

Due to constraint of nut: $\delta X_{BT} = \delta X_{ST}$ and since $L_S = L_B$

$$\epsilon_{XBT} = \epsilon_{XST}$$

$$\therefore \frac{\sigma_{XBT}}{30 \times 10^6} + 2.6 \times 10^{-4} = \frac{\sigma_{XST}}{10 \times 10^6} + 4.8 \times 10^{-4} \rightarrow \sigma_{XBT} = 3\sigma_{XST} + 6.6 \times 10^3 \quad \text{..... (2)}$$

$$\text{From equilibrium, } \sigma_{XST} A_S + \sigma_{XBT} A_B = 0 \rightarrow \sigma_{XST} \frac{\pi}{4} \left[\left(\frac{3}{4}\right)^2 - \left(\frac{1}{2}\right)^2 \right] + \sigma_{XBT} \frac{\pi}{4} \left(\frac{1}{2}\right)^2 = 0$$

$$\therefore 5\sigma_{XST} = -4\sigma_{XBT} \text{ putting in eq. (2)}$$

$$5\sigma_{XST} = -4\left(3\sigma_{XST} + 6.6 \times 10^3\right) \rightarrow$$

$$\sigma_{XST} = -1.55 \times 10^3 \text{ lb/in}^2$$

$$\sigma_{XBT} = 1.9 \times 10^3 \text{ lb/in}^2$$

ANS.

STRESS in Bolt: $\sigma_{XB} + \sigma_{XBT}$

$$= (3.36 + 0.19) \times 10^4 = 3.55 \times 10^4 \text{ lb/in}^2$$

STRESS in sleeve: $\sigma_{XS} + \sigma_{XST}$

$$= (-2.7 - 0.16) \times 10^4 = -2.86 \times 10^4 \text{ lb/in}^2$$

(5-42)

From equilibrium,

$$2\pi r t \sigma = \pi r^2 p \quad t = \text{thickness of balloon}$$

$$\sigma = \frac{P}{2} \left(\frac{r}{t} \right) \quad (1)$$

Because wall vol. of balloon is constant, (i.e. incompressible)

$$4\pi r_0^2 t_0 = 4\pi r^2 t$$

$$t = \left(\frac{r_0}{r} \right)^2 t_0 \rightarrow \left(\frac{r}{t} \right) = \left(\frac{r}{r_0} \right)^2 \frac{r_0}{t_0} \quad (2)$$

If we assume rubber to be linearly elastic

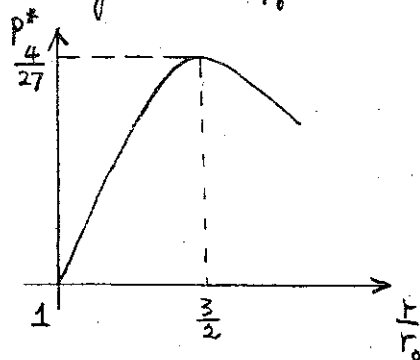
$$\frac{\sigma}{2} = E \epsilon = E \left(\frac{2\pi r - 2\pi r_0}{2\pi r_0} \right) = E \left(\frac{r}{r_0} - 1 \right) \quad (3)$$

substituting (2) into (1),
and equating to (3)

$$\frac{\sigma}{E} = \frac{r_0}{2E} P \left(\frac{r}{r_0} \right)^2 \left(\frac{r}{t_0} \right) \frac{1}{r_0} = 2 \left(\frac{r}{r_0} - 1 \right)$$

$$\text{Let } \frac{r}{r_0} = x$$

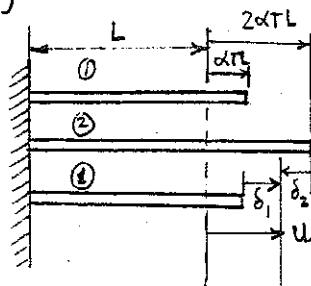
$$\therefore \frac{r_0 P}{2E t_0} x^3 = 2(x-1) \rightarrow P^* = \frac{x-1}{x^3} \quad P^* = \frac{P r_0}{4E t_0}$$

Plotting P^* vs $\frac{r}{r_0}$ To find location of max P^*

$$\frac{dP^*}{dx} = 0 = \frac{d}{dx} \left(\frac{1}{x^3} - \frac{1}{x^2} \right) = -\frac{2}{x^3} + \frac{3}{x^4} = \frac{-2x+3}{x^4}$$

$$\therefore x = \frac{3}{2}$$

(5-43)



From geometry:

Rods are constrained to move the same, so

$$u_1 = u_2 = u \quad (1)$$

From equilibrium: $\sum F = 0$ for rigid body

$$F_2 = 2F_1 \quad (2)$$

In terms of stresses: $\sigma_2 A = -2(\sigma_1 A)$

$$\therefore \sigma_2 = -2\sigma_1 \quad (3)$$

Where (-) sign accounts for the fact that the stresses are of opposite sense

Stress-strain-temperature relations: $u_1 = \alpha TL + \delta_1 = \alpha TL + \frac{F_1 L}{AE} \quad (4)$

$$u_2 = 2\alpha TL - \delta_2 = 2\alpha TL - \frac{F_2 L}{2AE} \quad (5)$$

Combining (1), (2), (4), (5), we get $T = \frac{2F_1}{\alpha AE} \quad (6)$ When yielding begins, the stress in bar ① is σ_y

$$\therefore F_1 = A\sigma_y \quad (7)$$

Putting (7) into (6), the temperature at which yielding occurs is:

$$(a) \quad T_y = \frac{2A\sigma_y}{\alpha AE} = \frac{2\sigma_y}{\alpha E} = \frac{2\epsilon_y}{\alpha}$$

(b) Using (4) to obtain u at $T = T_y$

$$u = u_1 = \epsilon_y L = \left[\frac{\sigma_y}{E} + \alpha \frac{2\epsilon_y}{\alpha} \right] L = 3\epsilon_y L$$

(c) If the temperature is increased beyond T_y , the stress σ_1 in bar ① remains constant at σ_y and therefore, by reason of equilibrium the stress σ_2 in bar ② remains constant at $\sigma_2 = -2\sigma_y$. Thus the displacement δ_2 which is uniquely dependent on the stress σ_2 does not change as the temperature is increased beyond T_y . Using (5)

$$\text{we get } u = u_2 = 2\alpha(2T_y)L - \delta_2 = 2\alpha\left(4\frac{\epsilon_y}{\alpha}\right)L - \frac{2\sigma_y}{2AE}L = 7\epsilon_y L \quad (13)$$

(d) The deflection δ_1 of bar ① consists of an elastic part ($\epsilon_y L$) and an unknown plastic part $\delta_{1p} = \delta_{1E} + \delta_{1p} = \epsilon_y L + \delta_{1p}$. Using (13) and (4), we calculate δ_{1p}

$$\delta_{1p} = 7\epsilon_y L - \epsilon_y L - \alpha\left(\frac{4\epsilon_y}{\alpha}\right)L = 2\epsilon_y L$$

$$\therefore \text{Plastic strain } \epsilon_{1p} = \frac{\delta_{1p}}{L} = 2\epsilon_y \quad 5-27$$

(5-44)



assume all stresses are produced by tensile forces.

Initial tightening produces a strain in bolt

$$\epsilon_B = \frac{\sigma_B}{E_B}$$

For equilibrium in bolt: $\sigma_w A_w + \sigma_B A_B = 0$

$$\therefore \sigma_w = - \frac{\sigma_B A_B}{A_w} = - \frac{E_B E_B A_B}{A_w} = - \frac{(5 \times 10^{-4})(205 \times 10^3)(310)}{(625)}$$

$$\sigma_w = -50.8 \text{ MN/m}^2$$

From increase in temperature:

$$\epsilon_{BT} = \frac{\sigma_{BT}}{E_B} + \alpha_B \Delta T$$

$$\epsilon_{WT} = \frac{\sigma_{WT}}{E_w} + \alpha_w \Delta T$$

$$\epsilon_{ST} = \frac{\sigma_{ST}}{E_s} + \alpha_s \Delta T$$

From equilibrium of bolt:

$$\sigma_{BT} A_B + \sigma_{WT} A_w = 0 \rightarrow \sigma_{BT} = - \frac{\sigma_{WT} A_w}{A_B}$$

From equilibrium of washer

$$\sigma_{WT} A_w - \sigma_{ST} A_s = 0 \rightarrow \sigma_{ST} = \frac{\sigma_{WT} A_w}{A_s}$$

From constraint of bolt: $\delta_{BT} = \delta_{WT} + \delta_{ST}$ $\therefore \epsilon_{BT} L_B = \epsilon_{WT} L_w + \epsilon_{ST} L_s$

$$L_B \left\{ \frac{\sigma_{BT}}{E_B} + \alpha_B \Delta T \right\} = L_w \left\{ \frac{\sigma_{WT}}{E_w} + \alpha_w \Delta T \right\} + L_s \left\{ \frac{\sigma_{ST}}{E_s} + \alpha_s \Delta T \right\}$$

$$L_B \left\{ - \frac{\sigma_{WT} A_w}{E_B A_B} + \alpha_B \Delta T \right\} = L_w \left\{ \frac{\sigma_{WT}}{E_w} + \alpha_w \Delta T \right\} + L_s \left\{ \frac{\sigma_{WT} A_w}{E_s A_s} + \alpha_s \Delta T \right\}$$

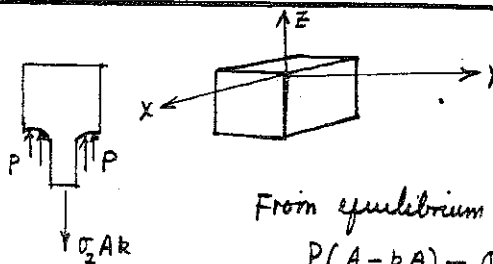
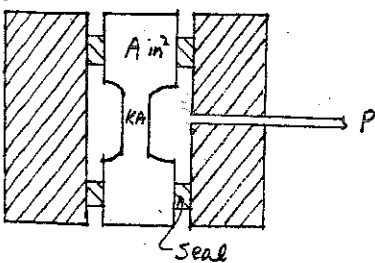
$$\sigma_{WT} \left\{ - \frac{L_s A_w}{E_s A_s} - \frac{L_w}{E_w} - \frac{L_B A_w}{E_B A_B} \right\} = \left\{ L_w \alpha_w + L_s \alpha_s - L_B \alpha_B \right\} \Delta T$$

$$\begin{aligned} L_w &= 1.5 \text{ mm}; L_s = 150 \text{ mm}; L_B = 151.5 \text{ mm} & \alpha_w &= \alpha_B = 10.5 \times 10^{-6} / ^\circ\text{C} \\ A_w &= 625 \text{ mm}^2 & \Delta T &= 95^\circ\text{C} & \alpha_s &= 21 \times 10^{-6} / ^\circ\text{C} \\ A_s &= 625 \text{ mm}^2 & E_w &= E_B = 205 \text{ GN/m}^2; & E_s &= 68 \text{ GN/m}^2 \\ A_B &= 310 \text{ mm}^2 \end{aligned}$$

$$\sigma_{WT} = -40.8 \text{ MN/m}^2$$

$$\text{TOTAL STRESS} = 40.8 + 50.8 = -91.6 \text{ MN/m}^2 \text{ COMPRESSION}$$

(5-45)



From equilibrium:

$$P(A - kA) - \sigma_z kA = 0$$

$$\sigma_z = P \left(\frac{1-k}{k} \right)$$

$$\text{also } \sigma_y A_y - P A_y = 0$$

$$\sigma_y = P$$

(5-45)(cont'd)

$$\sigma_x A_x - p A_x = 0 \rightarrow \sigma_x = p$$

material has yield stress $\sigma_Y =$

Mises criterion for principal axes:

$$\left[\frac{1}{2} \{ (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \} \right]^{1/2} = \sigma_Y$$

$$\therefore \left[\frac{1}{2} \{ (p-p)^2 + (p-p(\frac{1-k}{k}))^2 + (p(\frac{1-k}{k})-p)^2 \} \right]^{1/2} = \sigma_Y$$

$$p \left[1 - \left(\frac{1-k}{k} \right) \right] = \sigma_Y \rightarrow p = \frac{k}{2k-1} \sigma_Y$$

Max. shear-stress criterion: for $k > \frac{1}{2}$, $\sigma_z = \sigma_{\min}$; $\sigma_x = \sigma_y = \sigma_{\max}$

$$\frac{\sigma_{\max} - \sigma_{\min}}{2} = \frac{\sigma_Y}{2} \rightarrow p - p\left(\frac{1-k}{k}\right) = \sigma_Y \quad \text{or} \quad p = \frac{k}{2k-1} \sigma_Y \quad \text{Same as Mises criterion}$$

5-46 Residue Stress = $\sigma_x = 21 \text{ MN/m}^2$ (Tensile)

(a) altering temperature of plating (See Prob. 5-16)

Geometrical constraint requires $\epsilon_s = \epsilon_c$ (subscript 's' refers to steel
'c' refers to copper)

$$\epsilon_s = \alpha_s \Delta T \quad ; \quad \epsilon_c = \frac{1}{E} \sigma_x + \alpha_c \Delta T$$

$$\therefore \alpha_s \Delta T = \frac{1}{E} \sigma_x + \alpha_c \Delta T \rightarrow \Delta T = \frac{\frac{1}{E} \sigma_x}{(\alpha_s - \alpha_c)}$$

The value of $\alpha_c > \alpha_s$, σ_x is positive (tensile) therefore ΔT is negative and the plating needs to be done at below room-temperature to get rid of residue stress in copper!

(b) Desire to produce a strain so that the copper residue stress will be zero

$$\epsilon = \frac{\sigma_F}{E_{cu}} = \frac{21}{117 \times 10^3} = 1.79 \times 10^{-4}$$

$$\sigma_s = \epsilon E_s = 36.8 \text{ MN/m}^2$$

A uniaxial stress of 36.8 MN/m^2 is need.

(5-47) From eq. (5.9) for $P_o = 0, P_i = P$ $\sigma_r = -\frac{P[(\frac{r_o}{r_i})^2 - 1]}{(\frac{r_o}{r_i})^2 - 1}$; $\sigma_\theta = \frac{P[(\frac{r_o}{r_i})^2 - 1]}{(\frac{r_o}{r_i})^2 - 1}$; $\sigma_z = \frac{P}{(\frac{r_o}{r_i})^2 - 1}$

Rewriting in more convenient form:

$$\sigma_r = -\frac{P r_i^2 [(\frac{r_o}{r_i})^2 - 1]}{(r_o + r_i)(r_o - r_i)} = \frac{r_i [(\frac{r_o}{r_i})^2 - 1]}{(r_o + r_i)} \cdot \frac{-P r_i}{t}$$

$$\sigma_\theta = \frac{P r_i^2 [(\frac{r_o}{r_i})^2 - 1]}{(r_o + r_i)(r_o - r_i)} = \frac{r_i [(\frac{r_o}{r_i})^2 + 1]}{(r_o + r_i)} \cdot \frac{P r_i}{t} \quad t = r_o - r_i$$

$$\sigma_z = \frac{P r_i^2}{(r_o + r_i)(r_o - r_i)} = \frac{r_i}{(r_o + r_i)} \cdot \frac{P r_i}{t}$$

$$\frac{t}{r_i} = \frac{r_o - r_i}{r_i} = \frac{r_o}{r_i} - 1 \quad \lim_{\frac{t}{r_i} \rightarrow 0}, \quad \frac{r_o}{r_i} \rightarrow 1 \quad \text{or } r_o \rightarrow r_i$$

and $r \rightarrow r_i$

$$\lim_{\frac{t}{r_i} \rightarrow 0} \sigma_r = \frac{(\frac{r_o}{r_i})^2 - 1}{\frac{r_o}{r_i} + 1} \cdot \frac{-P r_i}{t} \rightarrow 0 \quad \text{stress constant across } t$$

$$\sigma_\theta = \frac{(\frac{r_o}{r_i})^2 + 1}{\frac{r_o}{r_i} + 1} \cdot \frac{P r_i}{t} \rightarrow \frac{P r_i}{t}$$

$$\sigma_z = \frac{1}{\frac{r_o}{r_i} + 1} \cdot \frac{P r_i}{t} \rightarrow \frac{P r_i}{2t}$$

5-48) From eq. (5.9), with $r_o = a, r_i = c, P_o = P, P_i = 0$

$$\left. \begin{aligned} \sigma_r &= -\frac{a^2}{a^2 - c^2} \left(1 - \frac{c^2}{r^2}\right) P \\ \sigma_\theta &= -\frac{a^2}{a^2 - c^2} \left(1 + \frac{c^2}{r^2}\right) P \end{aligned} \right\} \text{the maximum stress is } \sigma_\theta \text{ at } r=c$$

$$\text{at } r=c \quad \sigma_\theta = -\frac{a^2}{a^2 - c^2} \left(1 + \frac{c^2}{a^2}\right) P = -\frac{1}{(1 - \frac{c^2}{a^2})} 2P$$

$$\text{For } c \ll a, \quad \sigma_\theta \approx -2P$$

For a solid cylinder subjected to an external pressure P , the stresses σ_r and σ_θ are everywhere equal to $-P$ (hydrostatic pressure) so the effect of a small hole is to create a stress concentration

(5-49) Re-writing eq. (5-6) to eq. (5-8)

$$\left. \begin{aligned} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} + X_x &= 0 \\ \frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} + X_y &= 0 \\ \frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + X_z &= 0 \end{aligned} \right\}$$

$$\left. \begin{aligned} e_{xx} &= \frac{\partial u}{\partial x} & e_{xy} &= \frac{\gamma_{xy}}{2} = \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \\ e_{yy} &= \frac{\partial v}{\partial y} & e_{yz} &= \frac{\gamma_{yz}}{2} = \frac{1}{2} \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right) \\ e_{zz} &= \frac{\partial w}{\partial z} & e_{xz} &= \frac{\gamma_{xz}}{2} = \frac{1}{2} \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) \end{aligned} \right\}$$

Let x, y, z correspond to x_1, x_2, x_3 directions respectively

$$\begin{aligned} \frac{\partial \sigma_{11}}{\partial x_1} + \frac{\partial \sigma_{21}}{\partial x_2} + \frac{\partial \sigma_{31}}{\partial x_3} + X_1 &= 0 \\ \frac{\partial \sigma_{12}}{\partial x_1} + \frac{\partial \sigma_{22}}{\partial x_2} + \frac{\partial \sigma_{32}}{\partial x_3} + X_2 &= 0 \\ \frac{\partial \sigma_{13}}{\partial x_1} + \frac{\partial \sigma_{23}}{\partial x_2} + \frac{\partial \sigma_{33}}{\partial x_3} + X_3 &= 0 \end{aligned}$$

$$\therefore \sigma_{ij,i} + X_j = 0$$

Let u, v, w correspond to u_1, u_2, u_3 displacements

$$\begin{aligned} e_{xy} = e_{12} &= \frac{1}{2} \left(\frac{\partial u_2}{\partial x_1} + \frac{\partial u_1}{\partial x_2} \right) & e_{xx} = e_{11} &= \frac{1}{2} \left(\frac{\partial u_1}{\partial x_1} + \frac{\partial u_1}{\partial x_1} \right) \\ e_{yz} = e_{23} &= \frac{1}{2} \left(\frac{\partial u_3}{\partial x_2} + \frac{\partial u_2}{\partial x_3} \right) & e_{yy} = e_{22} &= \frac{1}{2} \left(\frac{\partial u_2}{\partial x_2} + \frac{\partial u_2}{\partial x_2} \right) \\ e_{xz} = e_{13} &= \frac{1}{2} \left(\frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1} \right) & e_{zz} = e_{33} &= \frac{1}{2} \left(\frac{\partial u_3}{\partial x_3} + \frac{\partial u_3}{\partial x_3} \right) \end{aligned}$$

$$\therefore e_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i})$$

$$\left. \begin{aligned} e_{xx} &= \frac{1}{E} \left[\sigma_{xx} + \nu \sigma_{yy} - \nu (\sigma_{xx} + \sigma_{yy} + \sigma_{zz}) \right] + \alpha (T - T_0) \\ e_{yy} &= \frac{1}{E} \left[\sigma_{yy} + \nu \sigma_{xx} - \nu (\sigma_{xx} + \sigma_{yy} + \sigma_{zz}) \right] + \alpha (T - T_0) \\ e_{zz} &= \frac{1}{E} \left[\sigma_{zz} + \nu \sigma_{xx} - \nu (\sigma_{xx} + \sigma_{yy} + \sigma_{zz}) \right] + \alpha (T - T_0) \\ \gamma_{xy} &= 2e_{xy} = \frac{\sigma_{xy}}{G} = \frac{2\sigma_{xy}(1+\nu)}{E} \\ \gamma_{yz} &= 2e_{yz} = \frac{2(1+\nu)\sigma_{yz}}{E} \\ \gamma_{xz} &= 2e_{xz} = \frac{2(1+\nu)\sigma_{xz}}{E} \end{aligned} \right\} \begin{aligned} e_{11} &= \frac{1+\nu}{E} \sigma_{11} - \nu (\sigma_{11} + \sigma_{22} + \sigma_{33}) + \alpha (T - T_0) \\ e_{22} &= \frac{1+\nu}{E} \sigma_{22} - \nu (\sigma_{11} + \sigma_{22} + \sigma_{33}) + \alpha (T - T_0) \\ e_{33} &= \frac{1+\nu}{E} \sigma_{33} - \nu (\sigma_{11} + \sigma_{22} + \sigma_{33}) + \alpha (T - T_0) \\ e_{12} &= \frac{\sigma_{12}}{E} \\ e_{23} &= \frac{\sigma_{23}}{E} \\ e_{13} &= \frac{\sigma_{13}}{E} \end{aligned}$$

$$\therefore e_{ij} = \frac{1+\nu}{E} \sigma_{ij} - \frac{\nu}{E} \delta_{ij} \theta + \alpha \delta_{ij} \Delta T$$

(5-50) Eq. (5-17) : Strain energy stored in a body

$$U = \frac{1}{2} \int_V (\sigma_x \epsilon_x + \sigma_y \epsilon_y + \sigma_z \epsilon_z + \tau_{xy} \gamma_{xy} + \tau_{yz} \gamma_{yz} + \tau_{zx} \gamma_{zx}) dV \quad \text{where } \tau_{xy} = \sigma_{xy}; \quad \gamma_{xy} = 2\epsilon_{xy} \text{ etc.}$$

$$= \frac{1}{2} \int_V (\sigma_{xx} \epsilon_{xx} + \sigma_{yy} \epsilon_{yy} + \sigma_{zz} \epsilon_{zz} + 2\sigma_{xy} \epsilon_{xy} + 2\sigma_{yz} \epsilon_{yz} + 2\sigma_{zx} \epsilon_{zx}) dV$$

Let x_1, x_2, x_3 correspond to x, y, z directions respectively.

$$U = \frac{1}{2} \int_V (\sigma_{11} \epsilon_{11} + \sigma_{22} \epsilon_{22} + \sigma_{33} \epsilon_{33} + 2\sigma_{12} \epsilon_{12} + 2\sigma_{23} \epsilon_{23} + 2\sigma_{31} \epsilon_{31}) dV$$

For $\sigma_{ij} = \sigma_{ji}, \epsilon_{ij} = \epsilon_{ji}$

$$2\sigma_{12} \epsilon_{12} = \sigma_{12} \epsilon_{12} + \sigma_{21} \epsilon_{21}$$

$$2\sigma_{23} \epsilon_{23} = \sigma_{23} \epsilon_{23} + \sigma_{32} \epsilon_{32}$$

$$2\sigma_{31} \epsilon_{31} = \sigma_{31} \epsilon_{31} + \sigma_{13} \epsilon_{13}$$

$$\therefore U = \frac{1}{2} \int_V (\sigma_{11} \epsilon_{11} + \sigma_{12} \epsilon_{12} + \sigma_{13} \epsilon_{13} + \sigma_{21} \epsilon_{21} + \sigma_{22} \epsilon_{22} + \sigma_{23} \epsilon_{23} + \sigma_{31} \epsilon_{31} + \sigma_{32} \epsilon_{32} + \sigma_{33} \epsilon_{33}) dV$$

$$= \frac{1}{2} \int_V (\sigma_{ij} \epsilon_{ij}) dV$$

(5-51) Strain energy stored in a body: eq. (5-17)

$$U = \frac{1}{2} \int_V (\sigma_x \epsilon_x + \sigma_y \epsilon_y + \sigma_z \epsilon_z + \tau_{xy} \gamma_{xy} + \tau_{yz} \gamma_{yz} + \tau_{zx} \gamma_{zx}) dV$$

From stress-strain relationship for an isotropic material, eq. (5.2), (5.17) becomes

$$U = \frac{1}{2} \int_V \left\{ \sigma_x \left[\frac{1}{E} \sigma_x - \frac{\nu}{E} (\sigma_y + \sigma_z) \right] + \sigma_y \left[\frac{1}{E} \sigma_y - \frac{\nu}{E} (\sigma_z + \sigma_x) \right] + \sigma_z \left[\frac{1}{E} \sigma_z - \frac{\nu}{E} (\sigma_x + \sigma_y) \right] \right.$$

$$\left. + \frac{\tau_{xy}}{G} \tau_{xy} + \frac{\tau_{yz}}{G} \tau_{yz} + \frac{\tau_{zx}}{G} \tau_{zx} \right\} dV$$

$$\therefore U = \int_V \left[\frac{1}{2E} (\sigma_x^2 + \sigma_y^2 + \sigma_z^2) - \frac{\nu}{E} (\sigma_x \sigma_y + \sigma_y \sigma_z + \sigma_x \sigma_z) + \frac{1}{2G} (\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2) \right] dV$$

From Prob. 5.7, omitting the temperature term,

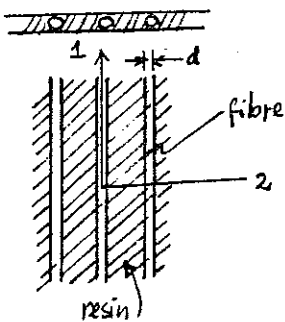
$$U = \frac{1}{2} \int_V \left\{ \left[\frac{(1-\nu)E}{(1+\nu)(1-2\nu)} \epsilon_x + \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_y + \epsilon_z) \right] \epsilon_x + \left[\frac{(1-\nu)E}{(1+\nu)(1-2\nu)} \epsilon_y + \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_z + \epsilon_x) \right] \epsilon_y \right.$$

$$\left. + \left[\frac{(1-\nu)E}{(1+\nu)(1-2\nu)} \epsilon_z + \frac{\nu E}{(1+\nu)(1-2\nu)} (\epsilon_x + \epsilon_y) \right] \epsilon_z + \frac{\tau_{xy}}{2(1+\nu)} \gamma_{xy} + \frac{\tau_{yz}}{2(1+\nu)} \gamma_{yz} + \frac{\tau_{zx}}{2(1+\nu)} \gamma_{zx} \right\} dV$$

Rearranging the terms, we obtain

$$U = \int_V \left\{ \frac{E(1-\nu)}{2(1+\nu)(1-2\nu)} (\epsilon_x + \epsilon_y + \epsilon_z)^2 - \frac{E}{(1+\nu)} \left[\epsilon_x \epsilon_y + \epsilon_y \epsilon_z + \epsilon_z \epsilon_x - \frac{1}{4} (\gamma_{xy}^2 + \gamma_{yz}^2 + \gamma_{zx}^2) \right] \right\} dV$$

(5-52)



(a) In the (1) direction the strains are equal in both the fibre and resin, so the loads carried by both are additive

$$P = P_f + P_r = \epsilon_{11} E_f A_f + \epsilon_{11} E_r A_r$$

$$\sigma_{11} = \frac{P}{A} = \epsilon_{11} \left[E_f \frac{A_f}{A} + E_r \frac{A_r}{A} \right] \quad \begin{array}{l} \text{subscript } f \text{ refers to fibers} \\ r \text{ refers to resin} \end{array}$$

$$E_{c1} = \frac{\sigma_{11}}{\epsilon_{11}} = E_f \left(\frac{A_f}{A} \right) + E_r \left(\frac{A_r}{A} \right) = (0.4)(5 \times 10^7) + (0.6)(5 \times 10^5) = 20.3 \times 10^6 \text{ psi}$$

If the load carrying capacity of the resin is ignored we have

$$E_{c1} = 20 \times 10^6 \text{ psi nearly equal to the exact solution}$$

In the (2) direction the loads are in series and the changes in length are additive

$$\delta l_c = \delta l_f + \delta l_r = \epsilon_f l_f + \epsilon_r l_r$$

$$\epsilon_f = \frac{\sigma}{E_f} \quad \epsilon_r = \frac{\sigma}{E_r} \quad \therefore \delta l_c = \sigma_{22} \frac{l_f}{E_f} + \sigma_{22} \frac{l_r}{E_r}$$

$$\text{dividing total length } l \text{ in (2) direction} \quad \frac{\delta l_c}{l} = \epsilon_{22} = \frac{\sigma_{22}}{E_f} \frac{l_f}{l} + \frac{\sigma_{22}}{E_r} \frac{l_r}{l}$$

$$\frac{1}{E_{c2}} = \frac{\epsilon_{22}}{\sigma_{22}} = \left[\frac{1}{E_f} \left(\frac{l_f}{l} \right) + \frac{1}{E_r} \left(\frac{l_r}{l} \right) \right]$$

ignoring the circular geometry of the fibres we may interpret l_f/l and l_r/l the relative volume fraction of fibre and resin

$$\frac{1}{E_{c2}} = (0.4) \left(\frac{1}{5 \times 10^7} \right) + \frac{0.6}{5 \times 10^5} = \frac{0.604}{5 \times 10^5}$$

or $E_{c2} = 8.3 \times 10^5 \text{ psi}$ If the extensibility of the fibres in the 2 direction is ignored, we obtained $E_{c2} = 8.35 \times 10^5 \text{ psi}$

(b) When 2 sheets are glued together, the new module in (1) and (2) directions will be equal and can be obtained by using the development for E_{c1} in part (a) i.e.

$$E_{c1,2} = E_{c1}(0.5) + E_{c2}(0.5) = [(20.3)(0.5) + (0.83)(0.5)] \times 10^6 = 10.565 \times 10^6 \text{ psi}$$

(c) No, isotropy is not assured because if composite is subjected to a stress in the direction 45° from axes (1) and (2), the composite would be under a state of shear stress referred to the (1) and (2) axes for which it will not have much resistance

(5-53)

$$e_1 = e_x, e_2 = e_y, e_3 = e_z, e_4 = \gamma_{xy}, e_5 = \gamma_{yz}, e_6 = \gamma_{zx}$$

$$\tau_1 = \sigma_x, \tau_2 = \sigma_y, \tau_3 = \sigma_z, \tau_4 = \tau_{xy}, \tau_5 = \tau_{yz}, \tau_6 = \tau_{zx}$$

from eq. (5-20)

$$e_1 = S_{11} \tau_1 + S_{12} \tau_2 + S_{13} \tau_3 + S_{14} \tau_4 + S_{15} \tau_5 + S_{16} \tau_6$$

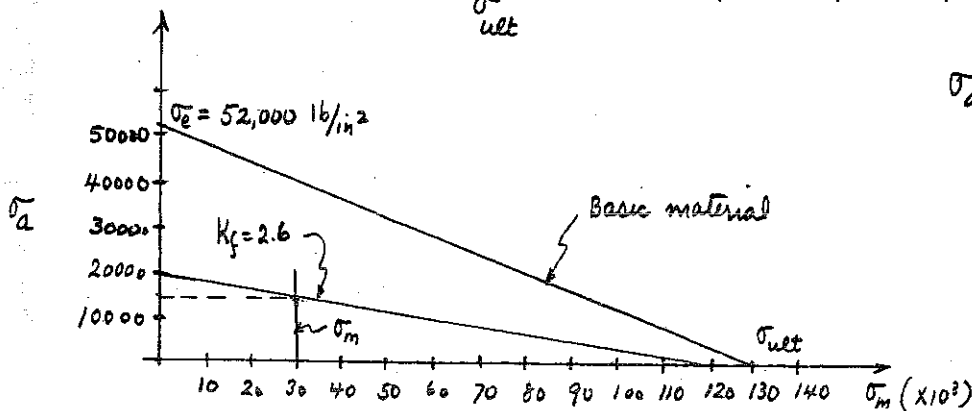
$$e_6 = S_{61} \tau_1 + S_{62} \tau_2 + S_{63} \tau_3 + S_{64} \tau_4 + S_{65} \tau_5 + S_{66} \tau_6$$

$$\therefore e_j = S_{ji} \tau_i \quad j = 1, \dots, 6$$

(5-54)

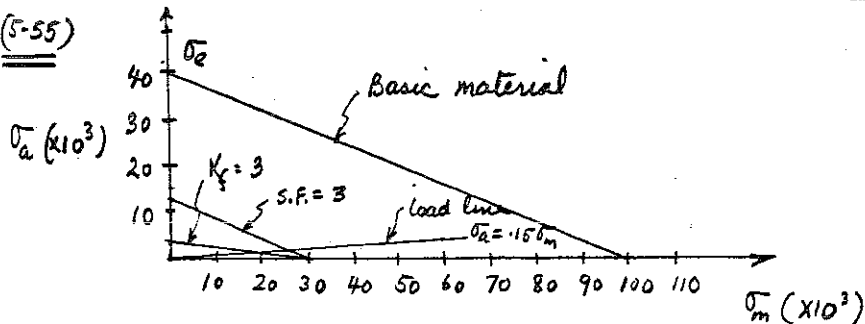
For a Ferrous metal

$$\frac{\sigma_e}{\sigma_{ult}} \cong 0.4 \quad (\text{infinite life, } \sigma_m = 0)$$



$$\begin{aligned} \sigma_a &= -\frac{2}{13} \sigma_m + 20,000 \\ &= -\frac{2}{13} (30,000) + 20,000 \\ &= 20,000 - 4,600 \\ &= 15,400 \text{ lb/in}^2 \end{aligned}$$

(5-55)



$$\sigma_a = -0.13 \sigma_m + 4.45 \times 10^3$$

$$0.15 \sigma_m = -0.13 \sigma_m + 4.45 \times 10^3$$

$$\therefore \sigma_m = \frac{4.45 \times 10^3}{2.8 \times 10^{-1}} = 1.6 \times 10^4 \text{ lb/in}^2$$

For a ferrous material

$$\frac{\sigma_e}{\sigma_{ult}} \cong 0.4 \quad (\text{infinite life } \sigma_m = 0)$$

$$\sigma_{ult} \cong \frac{\sigma_e}{0.4} = \frac{40,000}{0.4} = 10 \times 10^4 \text{ lb/in}^2$$

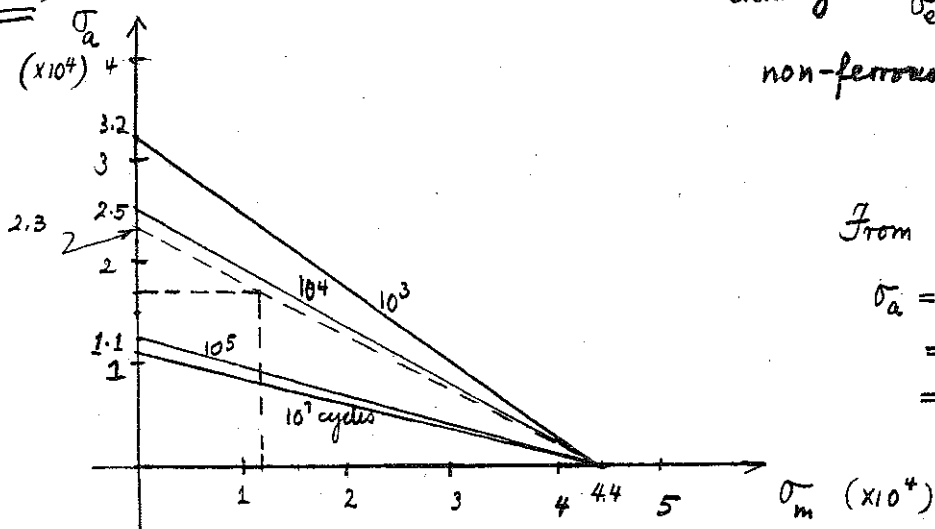
$$\sigma_a = 0.15 \sigma_m \quad (\text{load line})$$

$$\sigma_a = k \sigma_m + b$$

$$b = \frac{40 \times 10^3}{9} = 4.45 \times 10^3$$

$$k = \frac{-4 \times 10^4}{\frac{10^3}{3}} = -0.13$$

(5-56)



from Fig 5-42b $\sigma_e = 11000 \text{ psi} (10^7 \text{ cycles})$

non-ferrous metal $\frac{\sigma_e}{\sigma_{ult}} \approx 0.25$

$\sigma_{ult} = 44000 \text{ psi}$

From Fig 5-42b

$\sigma_a = 12000 \text{ psi}$ at 10^5 cycles
= 25000 psi at 10^4 cycles
= 32000 psi at 10^3 cycles

For $\sigma_a = 17000 \text{ psi}$ $\sigma_m = 12000 \text{ psi}$

number of cycle $\approx 1.2 \times 10^4$ cycles